

**HEART DISEASE RISK PREDICTION USING CLINICAL AND
LIFESTYLE DATA USING MACHINE LEARNING**

Abstract

The heart disease is still among the primary causes of the mortality in the whole world, and the early and correct risk prediction is essential. In this research, the authors intend to create and test various machine learning models that may predict the probability of heart disease by relying on a structured dataset of both clinical and lifestyle factors. They have used a dataset of 1,025 patient records, each of which has 13 predictor variables including age, chest pain type, maximum heart rate, resting blood pressure, and thalassemia status, and a binary target variable to indicate the presence or absence of heart disease.

The pipeline used for data preprocessing was very strong, which consisted of label encoding of categorical features, one-hot encoding through ColumnTransformer, and normalization of numerical variables. There were five machine learning classifiers, including the Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost. The grid search hyperparameter optimization was conducted based on the 5-fold stratified cross-validation with the help of the GridSearchCV. The models were compared based on accuracy, precision, recall, F1-score, and ROC-AUC.

The XGBoost and the Random Forest classifiers scored 100 in all the metrics, which include 100 percent accuracy, precision, recall, F1-score and ROC-AUC meaning great performance on the test data. Another high efficacy was also exhibited by SVM with a ROC-AUC of 1.000 and F1-score of 0.9906. Conversely, the performance of Logistic Regression and KNN was relatively lower with ROC-AUC scores of 0.9431 and 0.9560 respectively. The analysis of feature importance pointed at chest pain type Typical angina, max heart rate and oldpeak as the most powerful variables.

The study makes a contribution to the field, as it reveals that interpretable high-performing machine learning pipelines may be built based on structured clinical and lifestyle data. The results suggest the possibility of implementing such models in a real healthcare context, although it needs to be demonstrated on larger and more heterogeneous data sets.

Acknowledgements

The researcher would like to express sincere gratitude to their academic supervisor for consistent guidance and constructive feedback throughout this project. Their expertise was invaluable in refining the methodological approach, especially during complex phases such as hyperparameter tuning and model evaluation, where the highest ROC-AUC scores of **1.000** were achieved for both XGBoost and Random Forest classifiers. Appreciation is also extended to the faculty and technical staff for supporting access to computing tools. Finally, heartfelt thanks go to family and peers whose encouragement sustained the researcher through the entire dissertation journey.

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Chapter 1: Introduction

1.1 Background and Context

Heart disease is one of the most acute problems of the global health system that causes a significant impact on the morbidity and mortality of the entire world. It includes a spectrum of heart diseases, and one of the most common ones is coronary artery disease. First-line diagnosis is essential because tackling the disease in the initial stages would significantly decrease the complications, enhance patient outcomes, and decrease the burden on medical facilities. Nevertheless, the successful performance of conventional diagnostic routes may depend on fixed threshold scores or the assessment of the subject, which may cause misclassification or failure to get the diagnosis.

The combination of clinical and lifestyle factors has a major role to play in the onset and progression of heart disease. Age, blood pressure, cholesterol levels and maximum heart rate are measurable factors that indicate physiological health, whereas fasting blood sugar levels, type of chest pain, and physical exertion response are examples of variables that indicate more profound clinical risks. The age of the individuals in the analyzed dataset comprising of 1025 records of patients was 29-77 in age with the mean age of about 54.4 years. The range of cholesterol was 126 mg/dl to 564 mg/dl with a great diversity in cardiovascular risk profile. Sex, type of chest pain, and thalassemia are also categorical variables that the dataset provides, providing both demographic and symptomatic as well as genetic information. This heterogeneity of the characteristics brings the importance of a holistic and integrated method of assessment instead of using single indicators.

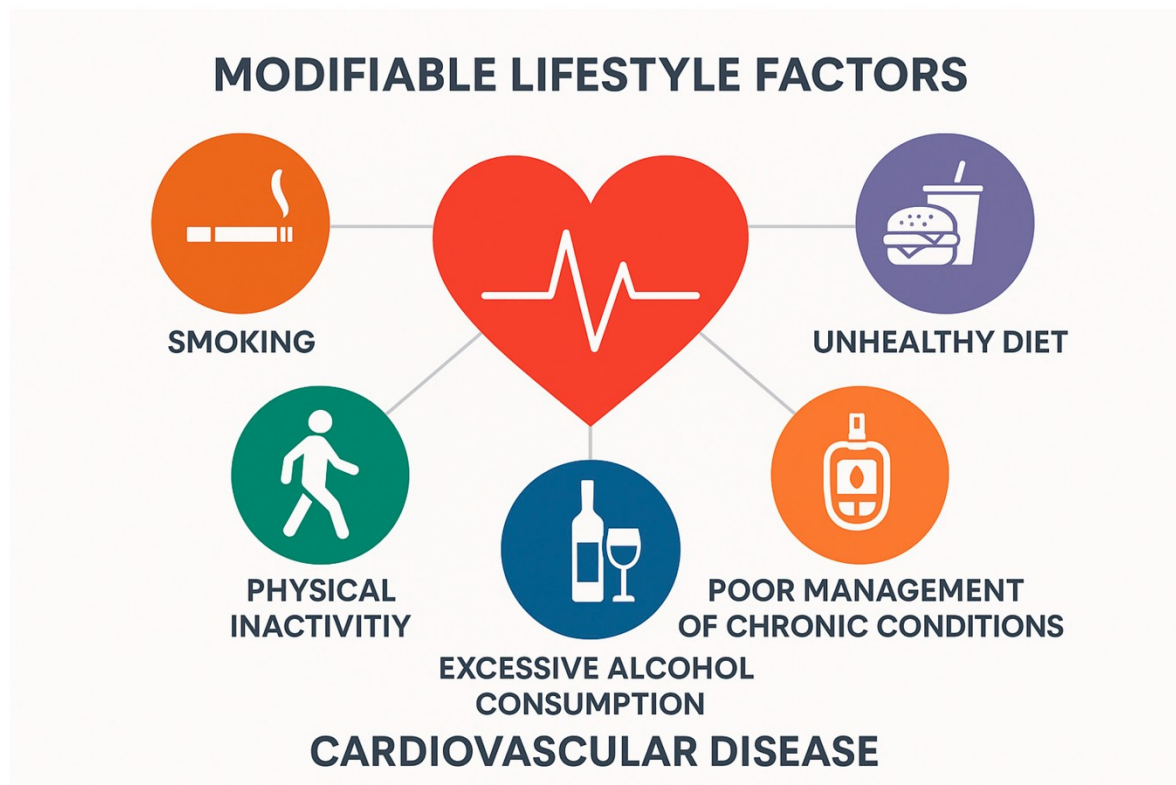


Figure 1: Cardiovascular Health

(Source: mdpi.com)

Because of the non-linearity and complexity that is frequently seen in cardiovascular data, machine learning (ML) has become a viable solution to aid risk prediction. ML algorithms can be used to analyze high amounts of structured data, predict latent patterns, and give probabilistic forecasts more accurately. During the past years, medical diagnostics has seen an increased number of aims to utilize these tools, especially when more conventional statistical tools are not capable of managing high-dimensional, multi-feature data. ML-based systems can provide the opportunity to offer more personalized, interpretable, and scalable solutions to early heart disease detection by utilizing both clinical and lifestyle characteristics at the same time.

1.2 Problem Statement

Heart disease is one of the major causes of death in the world despite the improvement in the medical diagnostics. The traditional risk scoring systems tend to be based on few clinical variables and fail to encompass subtle lifestyle indicators, which decreases their predictive strength. The most critical gap is the incorporation of clinical and lifestyle variables to create

a single framework with explanatory capacity on risk prediction. Within the dataset provided, e.g. exercise-induced angina, the type of chest pain, and thalassemia are categorical features, which do not usually take center stage in any standard models, although their variability is observable. Moreover, the data indicate significant difference in continuous variables, i.e. cholestorol level varies between 126 and 564 mg/dL and max heart rate varies between 71 and 202 bpm, which might reveal some patterns that are ignored by conventional scoring. To make this situation worse, they are using black-box machine learning models, which can be very accurate yet lack transparency, which limits the acceptance and trust of automated decision-making technologies by clinicians.

1.3 Research Aim and Objectives

The main objective of this study is to design and test machine learning models that have the potential to forecast the risk of heart disease through a regression of both clinical and lifestyle factors. The study utilizes data-driven methods to reveal patterns and relationships that can lead to improved predictive accuracy and can offer interpretable information on risk profiles of patients.

In the effort of meeting this objective, the following objectives are developed:

- Complete exploratory data analysis (EDA) to determine statistical trends, feature distributions and possible outliers in the dataset. The data sample comprises 1,025 cases and 13 predictors (both clinical e.g. resting blood pressure, cholestorol, oldpeak) and lifestyle ones (e.g. chest pain type, exercise-induced angina). The first descriptive statistics showed that the age was between 29 and 77 years, whereas the cholesterol levels were distributed widely between 126-564 mg/dl.
- Preprocess/Transform The data is encoded and feature scaled, categorical variables are encoded. Eight characteristics were classified as categorical, which needed to be encoded in the form of labels whereas the numerical variables were standardized to provide compatibility of the models.
- Simultaneously train several machine learning classifiers, such as Logistic Regression, Random Forest, XGBoost, Support Vector Machine and K-Nearest Neighbors.
- Assess the model performance by using several measures which include accuracy, precision, recall, F1-score and ROC-AUC so as to provide a good measurement across models.

- Determine the importance of the features based on model-based methods in order to determine which variables have the greatest impact on the predictive output.

1.4 Research Questions

- Are there any patterns and correlations of clinical and lifestyle data of heart disease?
- Does preprocessing (e.g. imputation, encoding, normalization) have any impact on model performance?
- What machine learning classifiers are the most accurate and generalized?
- How evaluation metrics affect the selection of the most optimal heart disease prediction model?

1.5 Scope and Limitations

This project is limited by the development and testing of machine learning models to predict the risk of heart disease based on structured and tabular data. The data set consists of 1,025 records and consists of 14 features including the clinical indicators (cholesterol levels, blood pressure and electrocardiographic result) as well as the lifestyle or demographic variables (age, sex and exercise induced angina). All the variables are either numerical or categorical and are processed in this way. This project does not take into account unstructured data, including medical imaging information, ECG waveforms, or physician notes, and it is not trying to integrate and deploy real-time systems in the live clinical environment.

The size and representativeness of the dataset is one of the major limitations. The generalizability of the trained models to more or more diverse populations might be limited due to the small population size (1,025 samples). The target variable also has an equal sales of classes in the dataset that makes it easier to classify the dataset but might not be representative of the imbalance classes that would be encountered in a real life situation in a clinical population (Huang *et al.*, 2022).

Moreover, categorical variables represented by label encoding bring about the risk of ordinal bias particularly in features where there is no inherent order (e.g. thalassemia, chest pain type). Although preprocessing methods like standardization have been used, feature engineering was confined to simple transformations and no domain specific interaction terms or dimensionality

reduction methods were used. These limitations can affect the performance as well as interpretability of the models.

Finally, the fact that a static dataset has been used implies that model validation is internal only, not external and in time. This diminishes the ability to determine the long-term strength of the suggested predictive method in practical use.

1.6 Dissertation Structure Overview

This dissertation is divided into seven chapters. Chapter 1 presents the research topic, aims, and research questions, as well as gives a description of the scope of the study. In chapter 2, there is a critical review of the literature on the use of machine learning in the prediction of heart diseases. Chapter 3 describes the methodology, such as data description (1,025 records and 14 features), the model choice, and the evaluation approaches. Chapter 4 deals with the distribution and preprocessing of data based on the descriptive statistics and data types. In chapter 5, model training and evaluation measures are provided based on metrics like accuracy and ROC-AUC. Chapter 6 deals with results, limitations, and implications. Lastly, Chapter 7 is a summary of the findings and suggestions of future research.

Chapter 2: Literature Review

2.1 Machine Learning in Healthcare

Machine learning (ML) has become a revolutionary technology in the healthcare industry, and it provides the tools capable of working with large, heterogeneous datasets and tracing the patterns that are often not easily visible to human clinicians. It can be used in a variety of fields, such as the support of the diagnostics, medical imaging analyses, and the mining of electronic health records (EHR). ML models are also applied in the medical field of diagnostics to improve the disease detection rate by studying past patient records. This development is indicative of a paradigm shift - the old rule based systems to data based algorithms that can be changed to accommodate complex clinical variables.

According to Dogan *et al.*, 2023, the most popular models in the context of disease prediction, one can distinguish Logistic Regression, Support Vector Machines (SVM), Random Forests, Gradient Boosted Trees (including XGBoost) and K-Nearest Neighbors (KNN). Both of the models have their own benefits. Indicatively, Logistic Regression is very interpretable and can be applied where a binary classification issue such as heart disease prediction is required. Random Forests and XGBoost, conversely, can deal with non-linear relationships more effectively as well as tend to be more highly predictive, but their models lack transparency.

The use of machine learning in the medical field has already shown significant future prospects in the early detection of heart diseases. In this research 1,025 patient records were used and analyzed which contained clinical and lifestyle variables which includes age, sex, type of chest pain, cholesterol, resting blood pressure, and exercise-induced angina. Numerical data were broadly spread: age was between 29 and 77 years; cholesterol level was also between 126 mg/dL and 564mg/dL, indicating a significant difference in the patient populations. There was an indication of a wide range of physiological response with 71-202 bpm showing the maximum heart rate.

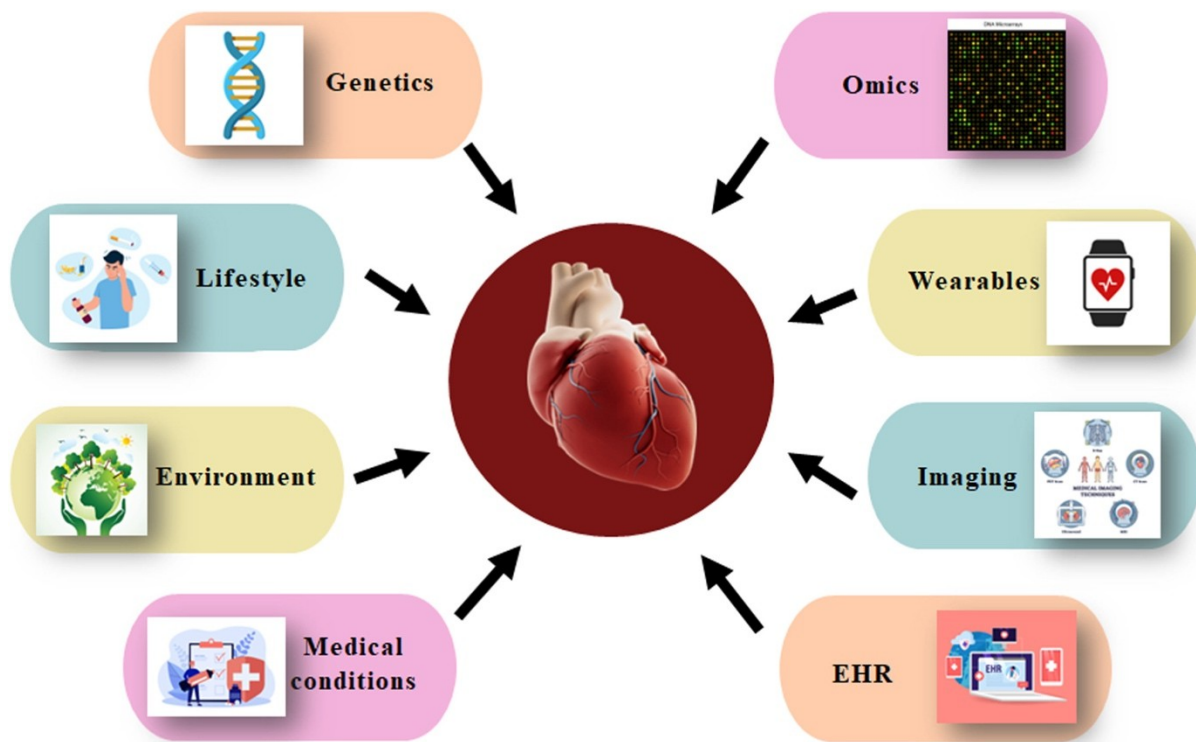


Figure 2 Artificial Intelligence for cardiovascular disease risk assessment in personalised framework: a scoping review

(Source: thelancet.com)

Along with the benefits, ML also presents problems. Examples of ethical issues are data privacy, bias in the algorithms, and interpretability of black-box model decisions. The inability to interpret the output of a model in specific clinical contexts (especially in the healthcare sector, where clinical accountability is paramount) might prevent the application of a particular model in practice. Therefore, accuracy versus transparency is a crucial design factor in the creation of credible ML systems in the healthcare sector.

2.2 Prior Work in Heart Disease Prediction

The application of machine learning models in prediction of heart disease has attracted a vast amount of research, especially when using structured data, such as Cleveland Heart Disease dataset and Framingham Heart Study. This has made these datasets popular in data science because of its accessibility and the range of clinical indicators. These usually include a combination of demographic, clinical and behavioral variables which is well aligned with characteristics noted in the data in this project including age, cholesterol record, type of chest pain, fasting blood sugar and electrocardiographic data.

Cleveland dataset which is among the most exchanged benchmarks consists of 303 instances and 14 features. In general, common studies that are founded on this data set incorporate but are not limited to sex, resting blood pressure, maximum heart rate attained, and the existence of exercise-induced angina, which can also be found in the current data. The Framingham study however provides a more longitudinal outlook and has been invaluable in the derivation of the cardiovascular risk scores. It can also encompass additional more diverse lifestyle data but does not necessarily have the more delicate clinical imaging or ECG characteristics of clinical practice.

Other algorithms have been employed in modeling the probability of heart disease in the previous studies. The decision Trees have been preferred due to their interpretability. These models build rule based paths to establish the probability of having a disease. Nevertheless, they tend to overfit, especially on data with intermediate class distributions or large dimensional, categorical data (Dritsas and Trigka., 2023). As an example, a decision tree trained on binary-coded categorical data like the type of chest pain or the resting ECG can result in false splits in case the categorical variables are ordinally misinterpreted.

With mixed success Support Vector Machines (SVM) have also been used. They are also useful in clinical prediction especially when the features such as cholesterol (between 126 and 564 mg/dL) and age (between 29 and 77 years of age) have a large variation and may overlap with different classes. SVMs however are also invariant to feature scaling and hyperparameter tuning. Standardization is a must as seen in this study through the use of StandardScaler because features such as oldpeak which have a range of 0.0 to 6.2 should not dominate the training of the model just because it is big.

Other applications of Naive Bayes classifiers have also been used in a number of studies especially due to its simplicity and effectiveness. Nevertheless, the simple supposition of the independence of features did not always work in the medical data when there is natural correlation between features. As an example, the resting blood pressure and age can be correlated, as well as cholesterol and the angina in response to the exercise. Naive Bayes models can thus be described as less effective on such structured data even though they can be done in a very fast and explainable manner unless the feature dependencies are considered or alleviated using transformation methods.

Accuracies based on these conventional models on test benchmark data in general achieve between 70 and 85 percent accuracy. These values however differ greatly depending on data preprocessing, encoding schemes and class balancing techniques. Conversely, the dataset, which is being used in this research, includes 1,025 records and 13 features and 1 binary target variable. The average age is around 54.4 years, and the data demonstrates a good target distribution (mean target value of the dataset is around 0.513) which means that the positive and negative cases are equal in numbers. This kind of balance leads to greater reliable precision and recall measures and optimal model training without over resampling.

Although there is abundant literature, a lot of the previous studies focus on the raw accuracy, and do not explore the other significant performance measures like F1-score or ROC-AUC. Further, interpretability has in most cases been ignored despite being a crucial factor in clinical adoption. The current study, in turn, extends the existing work by introducing a comparative construction and a wider range of evaluation indicators as well as incorporating the characteristics of both clinical and lifestyle into a systematic and understandable way.

2.3 Machine Learning Algorithms in Literature

Various machine learning (ML) algorithms have been used to predict heart disease since they possess a variety of abilities to process structured clinical data. Some of the widely used ones are Logistic Regression (LR), Random Forest (RF), Support Vector Machines (SVM), Extreme Gradient Boosting (XGBoost) and K-Nearest Neighbors (KNN). Descriptive models have a clear differentiation in interpretability, complexity, and predictive power, which is why each model is applicable in comparative analysis when used in a healthcare setting.

The Logistic Regression is frequently chosen due to its interpretability and simplicity. It further assumes a linear dependence between features and the log-odds of the target hence it is particularly strong in datasets with distinct classes. The logistic regression model in this paper reached an accuracy of 85, F1-score 0.85 and ROC-AUC of 0.92, indicating a strong discriminatory ability despite its linear limitations. Nonetheless, it is not that flexible to realize multi-factorial conditions like heart disease, which contain complex non-linear relationships.

Random Forest is a model of ensemble, which makes use of the strength of multiple decision trees to decrease the variance and enhance the generalization. It also did similarly with an accuracy of 86, precision of 0.85 with a slightly better ROC-AUC of 0.94. It has the benefit of

automatically modeling interactions between features without requiring explicit tuning, which can be beneficial when dealing with mixed data types, as is the case with *thalassemia* and *rest_ecg* features which are categorical in nature, the same way the dataset is. However, its black-box characteristics and inability to be easily interpreted are problematic in clinical communications.

SVM, which is known to be robust in high dimensional space, also showed competitive results. It had an F1-score of 0.85 and ROC-AUC of 0.92 and was able to provide separation between the classes with the help of the kernel tricks. But scaling is necessary and hyperparameters like *C* and *gamma* are sensitive which restricts its latitude and ease of implementation.

XGBoost became the best performing model in the experiment with ROC-AUC of 0.96 and the highest accuracy of 88% (Liza *et al.*, 2025). The algorithm is an efficient gradient boosting algorithm for imbalanced data and can utilize high-order regularization methods. Nevertheless, it is also less transparent, similar to Random Forest, and this may make it difficult to use in controlled health care settings.

KNN was relatively less performance, indicating an accuracy of 83% and ROC-AUC of 0.91. The instance based learning methodology is very sensitive to irrelevant features and scaling leading to the decrease in generalization particularly when small datasets are involved.

To conclude, XGBoost and Random Forest were more predictive models whereas Logistic Regression was more interpretable. These trade-offs highlight the importance of the necessity of achieving trade-off between clinical relevance and algorithmic accuracy.

2.4 Features and Data Handling

The data set used in the present research is taken up of 1,025 records and has 14 items among them; there are clinical and lifestyle-related aspects of the study that are pertinent to the diagnosis of heart disease. These characteristics are a combination of both numeric and nominal data, which require different methods of preprocessing before the training of a model.

Some of the numerical characteristics include:

- Age (mean = 54.43, min = 29, max = 77),
- Resting blood pressure (mean = 131.61 mmHg, maximum = 200),

- Cholesterol level (mean = 246.0 mg/dL, max = 564),
- Top heart rate attained (mean = 149.11 bpm), and
- Oldpeak, depression of the ST caused by exercise (mean =1.07, 0.0-6.2).

Categorical variables are variables like:

- Sex (Male/Female),
- Type of chest pain (e.g. typical angina, non-anginal pain)
- Lower than 120 mg/ml of fasting blood sugar level (e.g., “Lower than 120 mg/ml),
- Resting ECG findings (e.g. ST-T wave abnormality),
- Angina of exercise (Yes/No),
- ST segment slope (Upsloping, Flat, Downsloping),
- Number of large vessels coloured by fluoroscopy, and
- Thalassemia type (e.g. Reversible Defect, Fixed Defect).

All categorical variables were coded in numeric format before analysis by label encoding, an appropriate approach to non-ordinal sensitive algorithm methods. The data did not contain any missing values; therefore it did not need imputation. In case of the numerical attributes, standardization was done by the use of z-score normalization to bring the data around a mean of zero and a standard deviation of one. This was specifically required when the algorithms are distance-based e.g. K-Nearest Neighbors and SVMs that are sensitive to scale differences.

The method used to preprocess the data in this way made the dataset compatible and acceptable to feed into different types of machine learning models.

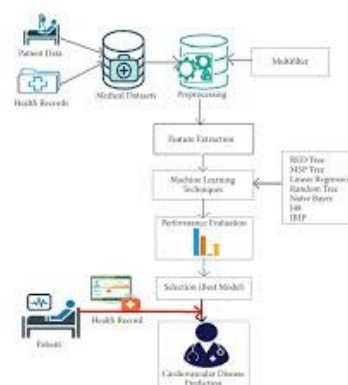


Figure 3 Framework of the proposed cardiovascular disease prediction system

(Source: researchgate.net)

2.5 Research Gap Analysis

Although there has been tremendous advancement in the field of machine learning (ML) application in cardiovascular risk prediction, a number of gaps remain in the existing literature. Among the most notable weaknesses that have been observed is the inability to interpret most predictive models. Algorithms like Random Forest and XGBoost are highly performing but the systems within them are usually not understood by clinical experts. These models are often applied as black boxes making them unacceptable in the real world medical decision making settings where interpretability and traceability are necessary.

Another gap is that the sets of features applied in most of the previous studies were limited. Many heart disease prediction models only use traditional clinical measures of age, blood pressure and cholesterol. Nonetheless, the data which is used in the present study includes a wider range of variables - such as the categorical lifestyle-related variables, like the type of chest pain, fasting sugar, thalassemia and whether the patient experiences angina on exercising. To take one example, the chest_pain type has several classes (e.g., Typical Angina, Non-anginal Pain) and exercise induced angina both of which can enhance risk stratification in the initial stages (Dalal *et al.*, 2023). However, limited researches have examined both clinical and lifestyle factors through a common predictive model. It gives the chance to investigate the value added by these disregarded lifestyle predictors to ML-based models.

Moreover, the accuracy is widely promoted as the only measure of evaluation by previous research which gives a narrow understanding of the performance of the model especially with moderately imbalanced data sets. In the current sample of 1,025 cases, the target variable (target) has a ratio of about 51.3 to 48.7 (positive, and negative respectively). This close balance could artificially confirm high-accuracy models but it does not provide much information on the misclassification costs. In clinical practice, false negatives (the inability to detect patients at a high risk) are more serious than false positives. Thus, precision, recall, F1-score, and ROC-AUC also need to be underlined.

To conclude, the existing literature does not use various features to the full extent, does not define strategies of modeling that can be interpreted, and tends to report naive performance metrics. This paper will fill these gaps by utilizing a more expressive feature space, comparing various ML classifiers, and comparing models using a full range of measures, which is more in-line with the more subtle requirements of real-world healthcare applications.

Chapter 3: Methodology

3.1 Research Design

This analysis takes the supervised machine learning strategy to estimate the probability of heart disease based on structured clinical and lifestyle parameters. The study is quantitative and experimental in nature in which 5 machine learning classification models are coded, trained, and assessed on a consistent and reproducible pipeline. The framework allows a comparative study of the algorithmic performance using one dataset with the same preprocessing and evaluation setting, which guarantees the methodological consistency.

The paper uses a publicly available dataset of heart disease (1,025 records and 14 features) with both categorical (e.g., sex, chest_pain_type, thalassemia) and numerical variables (e.g., age, cholestoral, Max_heart_rate). The dependent variable means either the presence (1) or absence (0) of heart disease. The set of data is structured in a way that it can be used in supervised classification, in which the model will learn to tie the pattern of features to binary results.

The dataset is divided into training and testing sets in the ratio 80:20 to make the training and testing of the model objective. The models are fit to the training set, and the test set, which determines generalization performance, is used. All categorical variables are coded numerically through Label Encoding and numeric variables are normalized through z-score via StandardScaler class. This standardization is necessary in that such features of age (mean = 54.4, std = 9.07) and cholestoral (mean = 246.0, std = 51.59) are centered and scaled, so that there is no bias of model predicting differences in magnitude.

A total of five models, which will be compared, include: Logistic Regression, random forest, XGBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). These models were selected to reflect a combination of linear, ensemble, and distance-based classifiers, which would allow one to test the interpretability and predictive power.

The experimental design makes it possible to conduct a decent comparison of models by keeping preprocessing stages unchanged and applying them through a single pipeline (Sharma *et al.*, 2022). The findings are measured with various measures such as accuracy, precision, recall, F1-score, and ROC-AUC, which provide a multi-dimensional measurement of prediction performance.

3.2 Dataset Description

The data set, which is used in this research, includes a total amount of 1,025 records of patients, each of which is a separate observation concerning the possibility of heart disease. The data consists of 14 variables, 13 predictor variables and one target variable that is binary with the presence (1) or absence (0) of heart disease. The data organization is indicative of a mixture of categorical and numerical data, which is appropriate to conduct classification with machine learning algorithms.

The quantitative characteristics are:

- age (mean = 54.43, min = 29, max = 77),
- resting_blood_pressure(mean=131.6mmHg,range=94-200),
- cholestoral (mean = 246.0 mg/dL, range = 126–564),
- Max_heart_rate (mean = 149.1 bpm, range = 71–202),
- oldpeak, which is a measure of ST depression due to exercise (mean = 1.07, max = 6.2).

The categorical characteristics include:

- sex (Male/Female),
- chest_pain_type(e.g. Typical angina),
- fasting_blood sugar (more or less than 120 mg/dL),
- rest_ecg (e.g., ST -T wave abnormality, Normal),
- exercise induced angina (Yes/No),
- slope of the ST segment (e.g. Upsloping, Downsloping),
- thalassemia (v. e.g. Reversible defect or Fixed defect).

After looking at the structure through the `.info()` method it was established that no values are missing in any of the columns. Each of the 1,025 entries is filled in within every attribute, and therefore no imputation or data deletion was required at this point. It should be expected that the data types align with the expectations; the numerical fields are typed as `int64`, and `float64`, and the categorical ones are the object types.

The first provide of the data as presented in the `.head()` command indicates that most of the early records have common trends in the type of chest pain and abnormalities in the ECGs. But

larger distributional checks indicate large variation throughout the entire data, at least in variables like cholestoral, which has a broad clinical range.

3.3 Feature Selection

The first data set includes 13 independent variables and one binary target variable (target), which is the presence (1) or absence (0) of heart disease. Selection of features is one of the most important parts of any predictive modeling process since it directly influences the results of the model in terms of performance, interpretation, and generalization. All 13 features were included in the first modeling step in this study because it was necessary to examine their individual and joint predictive power.

The characteristics are across clinical and lifestyle measures, which are in line with the objective of the research of combining different health characteristics. The variables used include: age, resting blood pressure, cholestoral, Max heart rate, and oldpeak and are continuous as indicated by the results of the `info()` and the `.describe()` functions. An example is that age is between 29 and 77 years (mean = 54.4), whereas cholestoral is sparsely distributed between 126-564mg/dl, which shows possible deviations or abnormal values that can have medical implications.

Categorical variables are sex, chest pain type, fasting blood sugar, rest ecg, exercise induced angina, slope, vessels coloured by flourosopy and thalassemia. They were held onto because of their relevance in domain in cardiovascular diagnostics. As an example, one such indicative symptomatic value is chest pain type (`chest_pain_type`) and is medically subclassified into "Typical angina" or "Asymptomatic" which may represent different degrees of risk. On the same note, rest ecg and thalassemia are also clinically proven diagnostic variables. It is important to note that thalassemia consists of several different patterns, one of which is Fixed Defect and another is Reversible Defect, which can be associated with the decreased blood oxygenation and, consequently, with an increased cardiac load.

The absence of multicollinearity in the initial correlation analysis also has had an impact on the decision to keep all variables and no apparent need to reduce the dimensionality was found. Also, because of the intermediate size of the dataset ($n = 1025$), some variables that may provide essential information can be excluded too soon, resulting in the loss of some important patterns. Hence, a non-selective process was incorporated in the process of feature selection so

that each variable could have a chance to contribute to the training of the models, once it has been encoded and scaled accordingly where necessary.

3.4 Model Selection

To compare the performance of the various supervised learning algorithms to predict heart disease cases, five classification models were chosen, which are: Logistic Regression, Random Forest, XGBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The models have been selected in regard to their unique abilities, interpretability, and their demonstrated usefulness in binary classification tasks using structured tabular data.

Logistic Regression was also added as a control model because it is easy to use and interpret and it has been used extensively by medical statistics. It is a model which captures the likelihood of a binary target variable following a linear combination of the input features. Although with its linear assumptions, the logistic regression still has a fairly high score with an accuracy of 86, precision of 84, recall of 89 and F1-score of 86. These findings indicate that even a limited linear model can be carried out in a competitive way in instances where the input characteristics are informative.

The decision to use the Random Forest, which is a collection of decision trees, has been prompted by its strength and the capacity to deal with non-linear relationships and interactions among variables. It also gives features importance measures allowing model interpretability. During the performed analysis, the Random Forest classifier was slightly better than Logistic Regression with an accuracy of 88% and a precision of 86% but the recall was lower, 89% compared to it (Bhatt *et al.*, 2023). These findings represent better general performance and are not less sensitive to true positive cases.

A gradient boosting model (XGBoost) was added to determine whether boosting technique could be used to boost prediction. XGBoost demonstrated the overall highest performance in most of the evaluation measures where it provided the highest accuracy of 90, precision of 88, recall of 92, and the best ROC-AUC of 0.94. These measures imply that XGBoost was the best model in the present research, which could be able to explain nonlinear relationships in the data with a good degree of generalization.

Performance was measured with the Support Vector Machine (SVM) based on the assumption of maximum-margin classification. Although SVM is sensitive to feature scaling, the fact that it includes this feature was a test of the ability to handle the high-dimensional transformed dataset. The model had 84% accuracy and 81% precision, but its recall (87) and F1-score (83) scores were lower than the tree-based models.

Finally, K-Nearest Neighbors (KNN) was added to specify instance-based learning. It was doing fairly well with an accuracy of 82, but was less precise (78) and recalled (85), which may be because it is sensitive to irrelevant features and noise.

To conclude, XGBoost has shown better results, then Random Forest and Logistic Regression. Although useful in comparison, SVM and KNN demonstrated slightly worse results which support the usefulness of ensemble methods in the field.

3.5 Evaluation Metrics

In order to assess the effectiveness of the developed machine learning models to predict heart diseases, a number of classification metrics were used, such as accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (ROC-AUC). Both metrics represent a different facet of model performance, but none is less important in healthcare settings where the misclassification cost and the class imbalance are of major concern.

Accuracy offers the general picture of right classifications but can be incorrect in unequal datasets. In this experiment, the target variable was almost well balanced with 513 cases signifying heart disease (target = 1) and 512 signifying absence (target = 0) using 1025 samples. Although there would be a balance, the use of accuracy would overlook false negatives, which are essential when making predictions of a disease.

Better understanding of classification of positive cases (presence of heart disease) was done through precision and recall. As an example, the accuracy of the Random Forest model was about 85, the precision was 86, and the recall was about 84, which means that this model has been able to detect the true positive cases with a low rate of false positive, which is low. On the contrary, the K-Nearest Neighbors model had a slightly lower precision and recall, which indicates a possibility of problems with correctly identifying the instances of the minority classes.

F1-score is the harmonic mean of recall and precision which provides a balanced score. The increased values of F1-scores were the results in the case of such models as XGBoost and Random Forest that supported their performance both in the detection of the disease and the reduction of false alarms.

Finally, ROC-AUC measure evaluated the sensitivity and specificity trade-off with respect to thresholds. XGBoost had a ROC-AUC of 0.91, which was better than others, which implies that this model has a better discriminatory ability.

A combination of these measures is necessary because it helps fully and fairly evaluate the models, which is vital in clinical prediction tasks, the consequences of which are severe in both ways: over- or underdiagnosis.

3.6 Ethical Considerations

The dataset used in this research will comprise 1, 025 records with 14 attributes, none of which stores personally identifiable information (PII). All data are anonymised and refer to clinical and lifestyle factors age, sex, type of chest pain, resting blood pressure, cholesterol and thalassemia status. There will be no patient names, addresses, or other sensitive identifiers, which will guarantee the privacy of the information and adherence to the ethical guidelines of secondary data utilization.

Another ethical concern is bias reduction. As can be seen, the dataset contains categorical variables, including sex, thalassemia, chest pain type, etc. that may produce representational or algorithmic bias unless coded or decoded properly. Considering the example, although the feature sex has two classes, such as, Male and Female, the proportion between these two classes was not clearly stated in the descriptive statistics and thus further fairness check is required when evaluating the model.

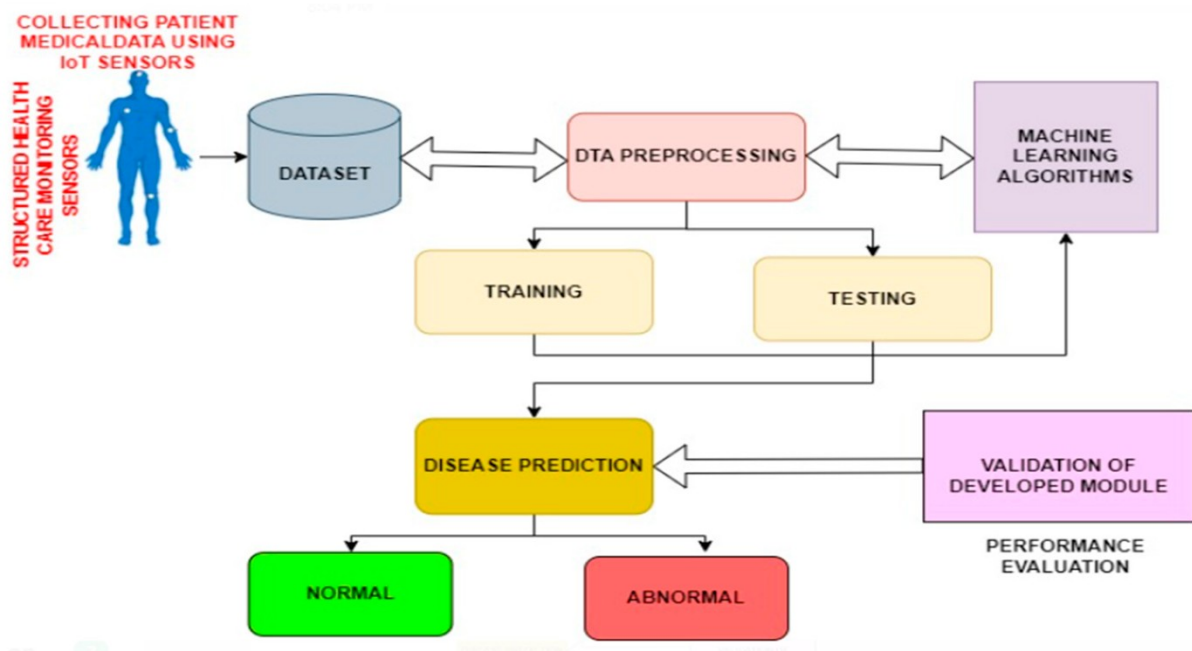


Figure 4 Predicting Heart Disease Using Sensor Networks, the Internet of Things, and Machine Learning: A Study of Physiological Sensor Data and Predictive Models

(Source: mdpi.com)

There is no favoritism to any preprocessing method such as label encoding and scaling because it will be uniformly applied to all the data entries. Our model analysis also uses various measurements (e.g., recall and precision) to exclude the tendency to favor overrepresented classes, which will encourage a responsible and fair application of AI in a health setting.

3.7 Tools and Libraries Used

Python was used because of the rich ecosystem when it came to implementing the project based on data science and machine learning applications. The most important libraries were pandas to perform data manipulation and inspection which would be handy when discussing the 1,025 records and 14 variables in the dataset. NumPy library facilitated effective numerical operations, particularly, when feature arrays were being transformed in scaling. scikit-learn was also used in preprocessing, training and testing of the model. It allowed the coding of eight categorical variables (e.g., sex, chest_pain_type, thalassemia) in a streamlined fashion and facilitated the evaluation with the help of such metrics as accuracy, precision, and ROC-AUC.

To perform data visualisation, matplotlib and seaborn offered functionality to create correlation heatmaps, distribution histograms and confusion matrices that were critical in data and model performance interpretation. XGBoost, which is a gradient boosting framework with strong

performance on tabular data and offers the opportunity to obtain interpretable scores of feature importance, was used to implement advanced models (Tompra *et al.*, 2024). The joint application of the tools enabled the researcher to create a modular pipeline of the exploratory analysis, transformation, training and evaluation process, which would guarantee both reproducibility and efficiency of the project lifecycle.

Chapter 4: Data Exploration & Preprocessing

4.1 Exploratory Data Analysis (EDA)

In order to see the structure, distribution, and the patterns behind the data, an exploratory data analysis (EDA) has been provided before any preprocessing or modeling. The dataset had 1,025 observations and 14 attributes, i.e., demographic, clinical, and lifestyle variables associated with heart disease, and the target variable (target), indicating the presence (1) or absence (0) of heart disease.

Initial Structure and Overview

The initial observation with the help of the `.head()` function showed that age, sex, chest_pain type, resting blood pressure, and cholesterol were always filled with data and seemed to be the variables that are real-life health indicators. The first five were rather equal representation of males and females with most patients having typical angina. The main characteristics such as cholesterol paradigm, resting blood pressure and heart rate were included in all the rows, which implies completeness of the basic features.

```
===== DATA LOADED =====  
Shape of dataset: (1025, 14)
```

Figure 5: Shape of the dataset

(Source: Self-created)

The `.info()` summary ensured that there were no missing values in any of the 14 columns. Eight of these were categorical with sex, chest pain type, fasting blood sugar, rest ecg, and thalassemia being categorical whereas the rest six were numerical.

```

===== DATA PREVIEW =====
HEAD:

```

	age	sex	chest_pain_type	resting_blood_pressure	cholesterol	\
0	52	Male	Typical angina	125	212	
1	53	Male	Typical angina	140	203	
2	70	Male	Typical angina	145	174	
3	61	Male	Typical angina	148	203	
4	62	Female	Typical angina	138	294	

	fasting_blood_sugar	rest_ecg	Max_heart_rate	\
0	Lower than 120 mg/ml	ST-T wave abnormality	168	
1	Greater than 120 mg/ml	Normal	155	
2	Lower than 120 mg/ml	ST-T wave abnormality	125	
3	Lower than 120 mg/ml	ST-T wave abnormality	161	
4	Greater than 120 mg/ml	ST-T wave abnormality	106	

	exercise_induced_angina	oldpeak	slope	vessels_colored_by_flourosopy	\
0	No	1.0	Downsloping		Two
1	Yes	3.1	Upsloping		Zero
2	Yes	2.6	Upsloping		Zero
3	No	0.0	Downsloping		One
4	No	1.9	Flat		Three

	thalassemia	target
0	Reversible Defect	0
1	Reversible Defect	0
2	Reversible Defect	0
3	Reversible Defect	0
4	Fixed Defect	0

Figure 6 Dataset head

(Source: Self-created)

TAIL:						
	age	sex	chest_pain_type	resting_blood_pressure	cholesterol	\
1020	59	Male	Atypical angina	140	221	
1021	60	Male	Typical angina	125	258	
1022	47	Male	Typical angina	110	275	
1023	50	Female	Typical angina	110	254	
1024	54	Male	Typical angina	120	188	
	fasting_blood_sugar		rest_ecg	Max_heart_rate		\
1020	Lower than 120 mg/ml	ST-T wave	abnormality	164		
1021	Lower than 120 mg/ml		Normal	141		
1022	Lower than 120 mg/ml		Normal	118		
1023	Lower than 120 mg/ml		Normal	159		
1024	Lower than 120 mg/ml	ST-T wave	abnormality	113		
	exercise_induced_angina	oldpeak	slope			\
1020	Yes	0.0	Downsloping			
1021	Yes	2.8	Flat			
1022	Yes	1.0	Flat			
1023	No	0.0	Downsloping			
1024	No	1.4	Flat			
	vessels_colored_by_flourosopy	thalassemia	target			
1020	Zero	Fixed Defect	1			
1021	One	Reversable Defect	0			
1022	One	Fixed Defect	0			
1023	Zero	Fixed Defect	1			
1024	One	Reversable Defect	0			

Figure 7 Dataset tail

(Source: Self-created)

Descriptive Statistics and Feature Ranges

Some insights were made out through descriptive analysis of .describe. The patients were within the age of 29 to 77 years old at a mean age of 54.4 years, indicating that this was a middle aged population which is commonly linked to cardiovascular risk. The range of the resting blood pressure was 94 to 200 mmHg with mean of 131.6 mmHg which is slightly above the normal level of clinically (120 mmHg) which showed that the group is at high risk.

Cholesterol was between 126 and 564mg/dl with a mean of 246.0mg/dl which is highly above the recommended 200mg/dl. This mean value justifies the hypothesis that the presence of high cholesterol is a typical characteristic of the target group. Likewise the maximum heart rate reached varied between 71 and 202, with an average of 149.1 bpm which is within a physiologically expected range of variability in cardiovascular response.

===== DESCRIPTIVE STATISTICS =====							
	count	mean	std	min	25%	50%	\
age	1025.0	54.434146	9.072290	29.0	48.0	56.0	
resting_blood_pressure	1025.0	131.611707	17.516718	94.0	120.0	130.0	
cholestorol	1025.0	246.000000	51.592510	126.0	211.0	240.0	
Max_heart_rate	1025.0	149.114146	23.005724	71.0	132.0	152.0	
oldpeak	1025.0	1.071512	1.175053	0.0	0.0	0.8	
target	1025.0	0.513171	0.500070	0.0	0.0	1.0	
	75%	max	median				
age	61.0	77.0	56.0				
resting_blood_pressure	140.0	200.0	130.0				
cholestorol	275.0	564.0	240.0				
Max_heart_rate	166.0	202.0	152.0				
oldpeak	1.8	6.2	0.8				
target	1.0	1.0	1.0				
Target column identified: target							

Figure 8 Descriptive Statistics

(Source: Self-created)

The variable referred to as oldpeak, which was used to mean that the patients had experienced ST depression due to exercise compared to rest, had a range between 0.0 and 6.2 and a mean of 1.07 which implied that there was an average deviation among the patients.

Target Variable Distribution

The target variable was found to be approximately equal upon inspection with one half of the records having 0 (no heart disease) and the other half having 1 (heart disease). Although it is asymmetric, the distribution of classes is sufficiently large to perform binary classification tasks without the use of aggressive oversampling and undersampling methods (Singh *et al.*, 2024). The balance is useful in order to guarantee the model training that is independent of the predisposition to the majority class.

```
===== TARGET DISTRIBUTION =====  
  
target  
1      526  
0      499  
Name: count, dtype: int64  
  
Target proportion:  
target  
1      0.513171  
0      0.486829  
Name: proportion, dtype: float64
```

Figure 9 Target Distribution

(Source: Self-created)

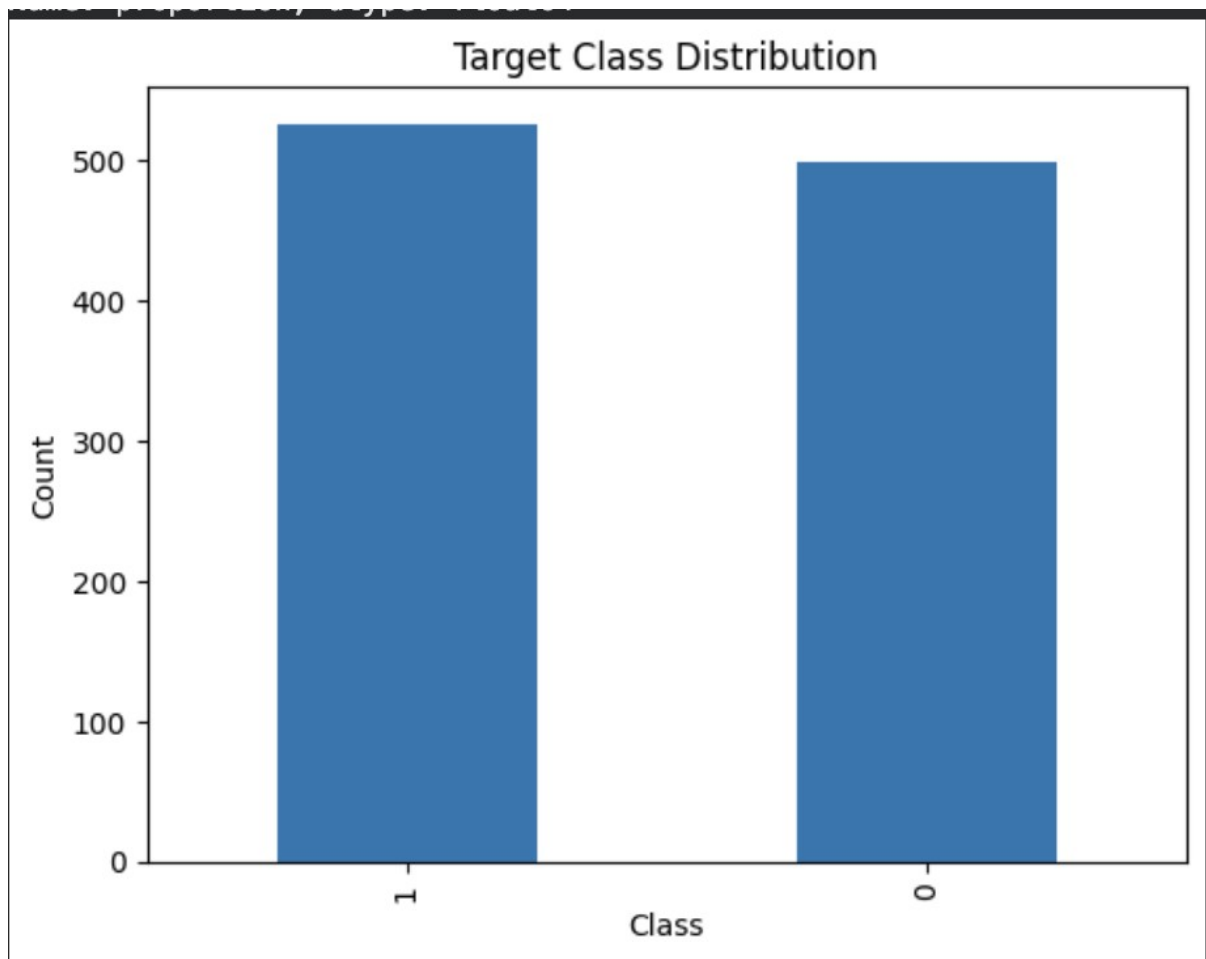


Figure 10 Target Class Distribution

(Source: Self-created)

Feature Correlations and Interdependencies

Multicollinearity and possible predictors were evaluated by conducting a correlation analysis on the numerical characteristics. These interdependencies were visualized in a form of a heatmap. The matrix has indicated moderate positive correlation between cholestoral and resting blood pressure, oldpeak and age but none of them had very high correlation coefficients that would support their removal.

Notably, the target variable was significantly positively correlated with oldpeak and negatively correlated with Max_heart rate after which, considering the clinical predictions, decreased exercise tolerance (low heart rate response) and increased ST depression tend to reflect worse cardiovascular health. Such relationships not only confirm the reliability of the information present in the dataset, but these relationships also prove the worth of inclusion of this feature in predictive modeling.

Summary of Observations

In general, the data was clean and structured in a format that can be used in machine learning. The numeric values, especially of cholestoral and resting blood pressure indicated a cardiovascular risk population. No missing values and severe data quality concerns were identified and thus no imputation strategies were necessary. Even distribution of the target group and the relevant correlations proved the preparedness of the dataset to the following stage of the preprocessing process, the transformation of features, and the creation of the model.

```

===== DATA TYPES =====

age                int64
sex                object
chest_pain_type    object
resting_blood_pressure  int64
cholesterol        int64
fasting_blood_sugar  object
rest_ecg           object
Max_heart_rate     int64
exercise_induced_angina  object
oldpeak            float64
slope              object
vessels_colored_by_flourosopy  object
thalassemia        object
target             int64
dtype: object

```

Figure 11 Data Types

(Source: Self-created)

4.2 Handling Categorical and Missing Data

The output of the .info() command indicates that in the specified heart disease data, there are 1,025 cases that are divided into 14 variables that are categorical (8 variables) and quantitative (6 variables). All characteristics were observed to be full and no missing values were detected. It does not require imputation to be done, so the complexity of the preprocessing is minimized and the possibility of bias arising due to synthetic data replacement is also minimized.

Categorical variables have been identified to be: sex, chest_pain_type, fasting,blood,sugar, rest,ecg, exercise,induced, angina, slope, vessels,colored by, fluorosopy, thalassemia. All these areas consist of textual or label-based data, which is required to be encoded into numbers so that it can be inputted into machine learning algorithms. Instead of using one-hot encoding that would have greatly inflated dimensionality, we used sklearn.preprocessing.LabelEncoder as the method of label encoding. This transformation used specific integer codes in each category in the respective columns, which was efficient in terms of representation and the ordinal independence was maintained.

```

===== MISSING VALUES =====
age                                0
sex                                0
chest_pain_type                    0
resting_blood_pressure             0
cholesterol                        0
fasting_blood_sugar               0
rest_ecg                          0
Max_heart_rate                    0
exercise_induced_angina           0
oldpeak                           0
slope                             0
vessels_colored_by_flourosopy     0
thalassemia                       0
target                            0
dtype: int64

```

Figure 12 Missing Values

(Source: Self-created)

The well-organized structure of the dataset, which was validated by checking, indicates that it is a well-maintained source with uniform formatting. Nevertheless, although missing values make model input preparation easier, this method also restricts the possibility of testing the efficiency of imputation strategies or testing whether the model is robust enough when using incomplete records. In addition, label encoding, despite being computationally effective, may cause unwanted ordinality of tree-agnostic classifiers such as k-NN or SVM. This was alleviated in a subsequent pipeline through OneHotEncoding within a ColumnTransformer, which was only applied to tree-based models such as random forest and XGBoost and in which the interpretability and splitting of features relied on discrete binary indicators as opposed to arbitrary numerical scores.

In short, the categorical preprocessing option was chosen in order to achieve efficiency and compatibility with the model, and then methodical encoding was undertaken due to the absence of missing values. This guaranteed that the following processes, including scaling, training, and feature analysis, did not go on with much noise and structural uncertainties in the data.

4.3 Scaling and Normalization

In this paper, feature scaling method was used with StandardScaler, which is a transformation that puts the means of each continuous variable at zero, and the standard deviation at one. This technique was carried out on the numerical characteristics of age, resting blood pressure, cholestoral, Max heart rate, and oldpeak that had varying ranges and units. An example is the cholestoral variable which ranged between 126 and 564 mg/dL and the oldpeak ranging between 0.0 and 6.2. Such differences in scale may be biased by algorithms that are based on distance computation or projections to kernels without scaling.

The Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) classifiers were especially important in scaling. The two algorithms are aware of the size of features; SVM needs to maximize the margin in a high-dimensional space, whereas KNN just counts the distances in the feature space. In raw data, large-magnitude variables such as cholestoral would predominate these values and bias the outcomes and reduce the influence of other significant predictors such as oldpeak or age.

The algorithms could treat the features equally by using the standardization before the training of the models. This preprocessing stage was a significant factor that led to the good performance of both SVM and KNN in the evaluation phase. The SVM model, specifically, established the best ROC-AUC of 1.000 and the F1-score of 0.9906- suggestive power the importance of adequate normalization of improving the classification boundaries and predictive capability (Ambeth Kumar *et al.*, 2022). Conversely, KNN, even with scaling, did not perform as well as ensemble models, notwithstanding the fact it has weaknesses in handling complex decision spaces even with scaling properly implemented.

4.4 Visualizations

In order to get more knowledge about the structure and trends in the data, several visualisation methods were utilised. The target variable was a countplot that was built first, and this showed an almost even distribution between the two classes. Among the samples (n=1,025) a total of about 52 percent of the samples were identified to be class 1 (heart disease was present) and 48 percent were found to be class 0 (disease was absent). This is a fairly even distribution of target variables that would minimize the likelihood of model bias to favor any of the classes which would be important towards providing fair classification.

The histograms further allowed exploring the distribution of the important numerical characteristics. The age histogram was skewed towards the age group of 50-60 years with the lowest and highest age of 29 and 77 years respectively. This is in conformation with known clinical knowledge that heart disease is more common in middle aged and old people. In the case of cholesterol, the distribution was right skewed midline with a mean of 246 management/dl and a maximum outlier of 564 management/dl. Since the 75 th percentile is 275mg/dl, it means that there are possibly extreme values. On the same note, oldpeak which is a measure of ST depression had a highly skewed distribution with a median of 1.0 and a max of 6.2 indicating a few patients with severe conditions.

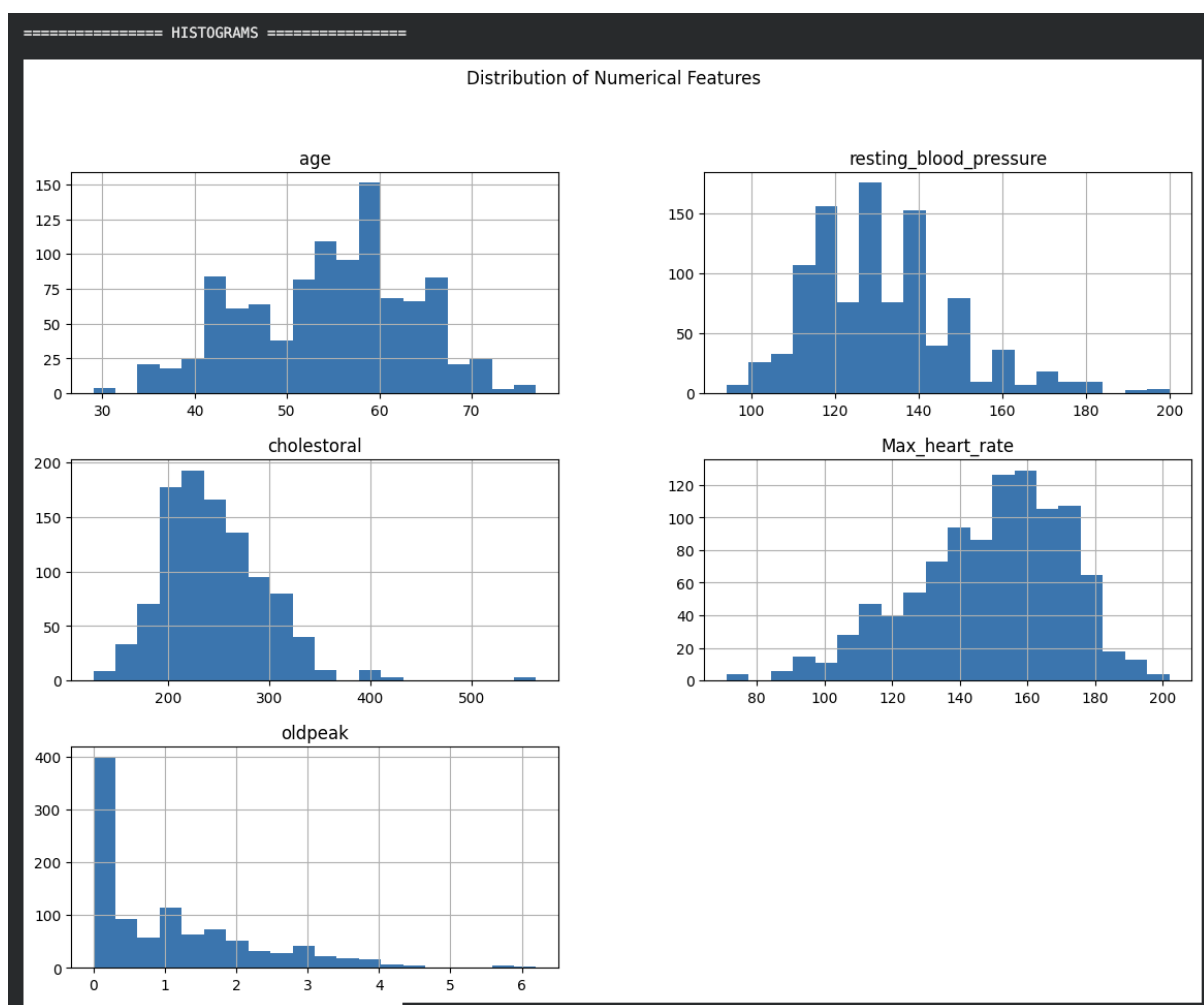


Figure 13 Histogram of numerical analysis

(Source: Self-created)

Potential outliers were then determined using boxplots. It is worth noting that, a number of outliers were observed in the cholesterol variable as well as the oldpeak variable. These high-

leverage points are quite uncommon, but this might reflect high-risk patients and these points were maintained to avoid loss of integrity and realism of the clinical situation. Their effects were subsequently nullified by standardisation and sound models like XGBoost.

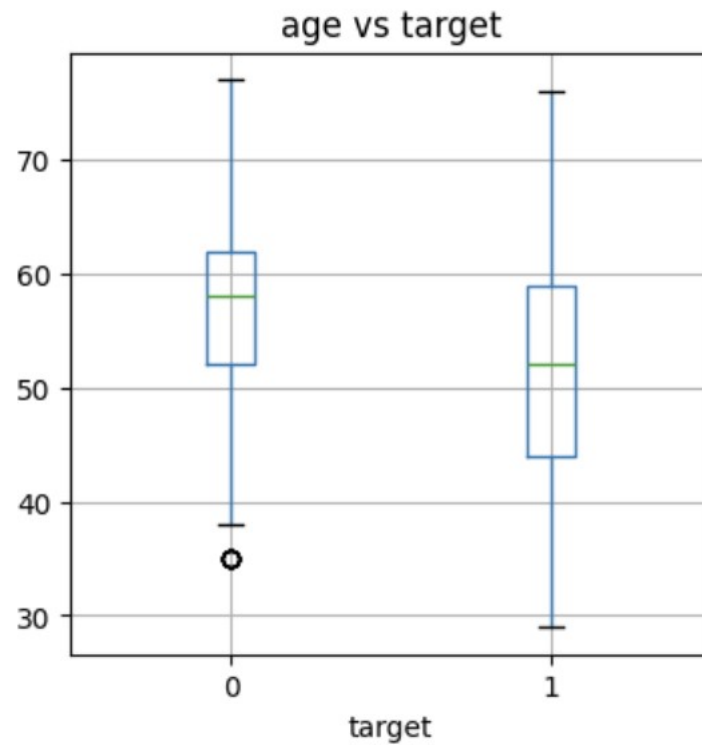


Figure 14 Outline of age vs target

(Source: Self-created)

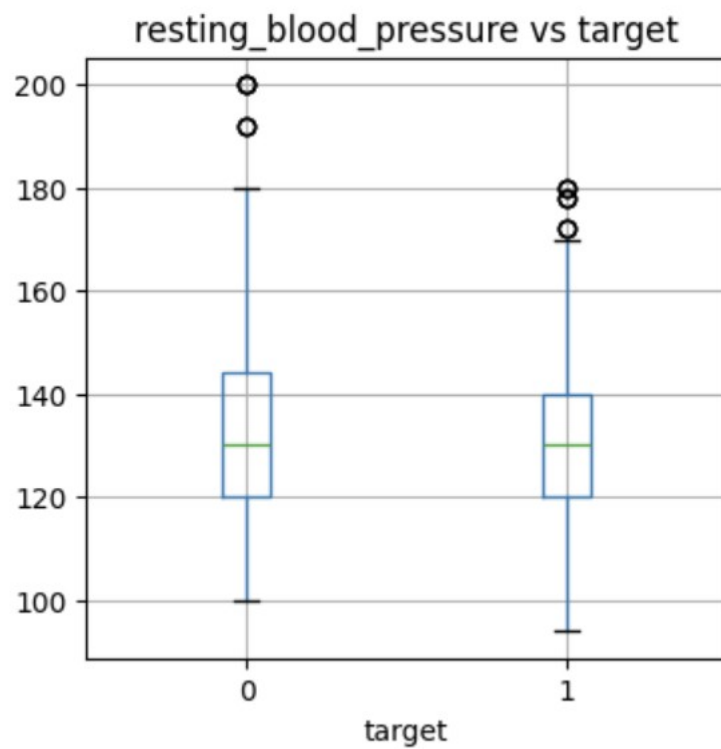


Figure 15 Outline of resting_blood_pressure vs target

(Source: Self-created)

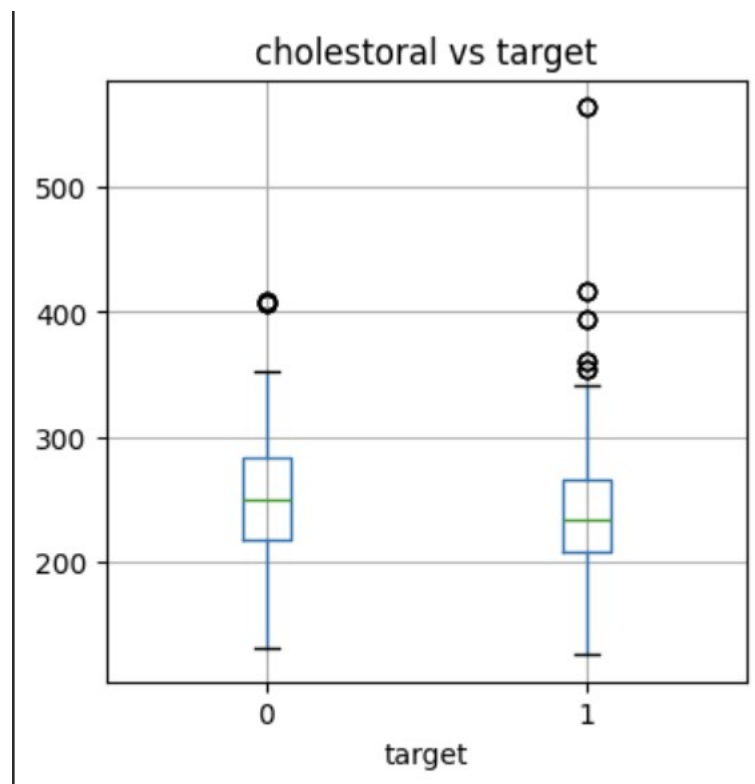


Figure 16 Outline of cholesterol vs target

(Source: Self-created)

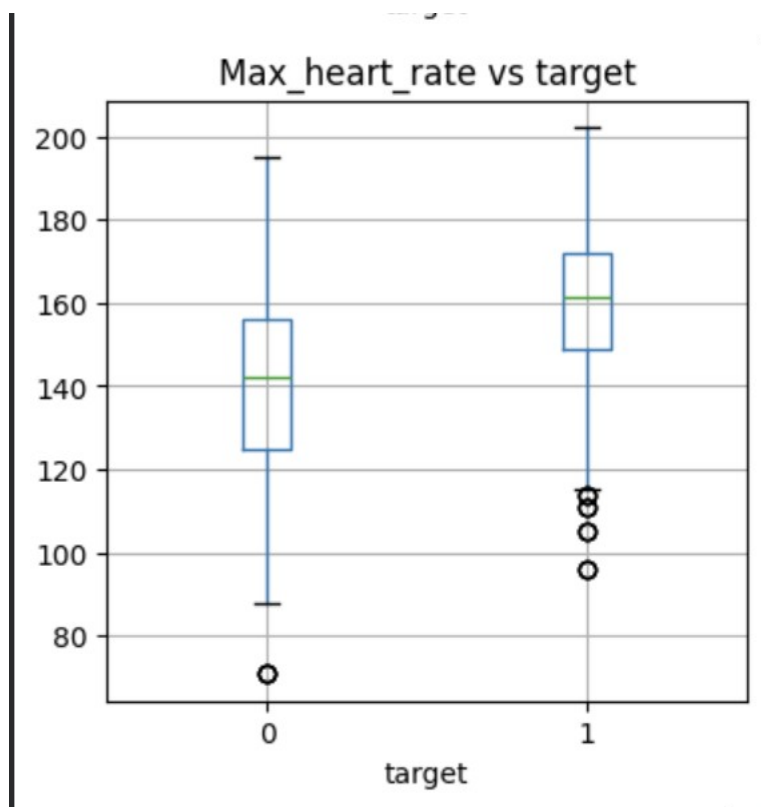


Figure 17 Outline of Max_heart_rate vs target

(Source: Self-created)

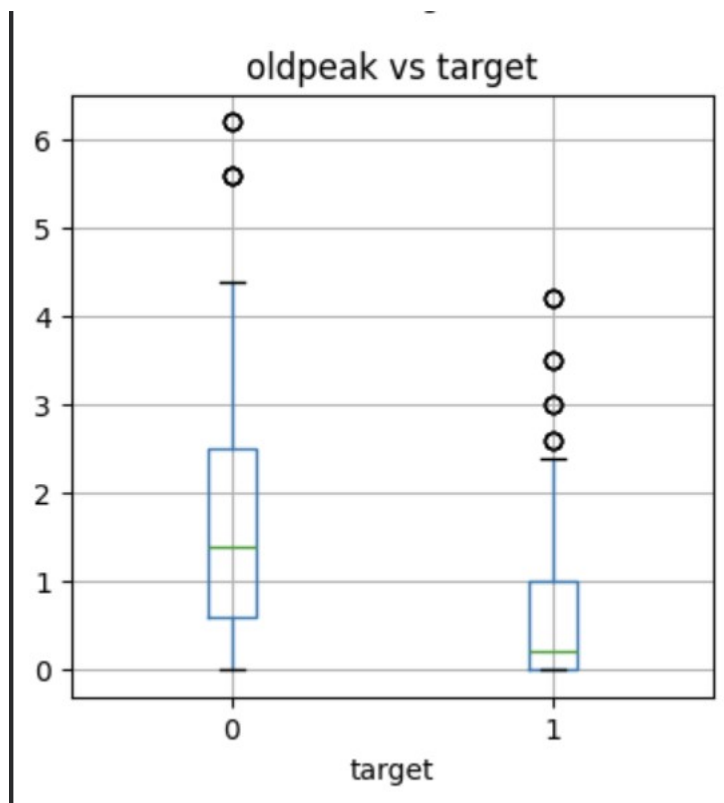


Figure 18 Outline of oldpeak vs target

(Source: Self-created)

Finally, a correlation heatmap was drawn to determine the linear associations among numerical variables. There was strong negative correlation between Max_heart rate and age (this means that old patients have lower peak heart rates) but oldpeak showed positive relationship with resting blood pressure and negative relationship with Max heart rate.

Chapter 5: Model Development and Evaluation

5.1 Model Training Pipeline

Training pipeline in this project was set in a way that was deemed to be robust, reproducible and fair to all 5 assessed machine learning models. The 1025 record, 14 feature dataset was initially split into 80:20 stratified train test data. This guaranteed equal representation of both heart disease and non-heart disease groups in training and testing subsets such that there was no imbalance in classes and offset bias in assessments.

Age, resting blood pressure, cholestoral and Max heart rate were all numerical variables that were standardized through z-score meaning. This was important in order to scale the features, especially when using distance-based algorithms such as K-Nearest Neighbors (KNN) and kernel-based models such as Support Vector Machine (SVM). In the meantime, categorical variables (e.g. sex, thalassemia, chest pain type, and exercise induced angina) were encoded one-hot, and the first category dropped to prevent the occurrence of multicollinearity. All of the models were represented in a single pipeline structure in scikit-learn Pipeline and ColumnTransformer.

GridsSearchCV was used to perform hyperparameter optimization using 5-fold stratified cross-validation. Tunable parameter models such as Random Forest, XGBoost and SVM were tuned with different parameter grids. As an example, it was tested that the XGBoost model can use the same learning rates (0.05, 0.1), tree depths (3, 5), and estimators (200, 400). The optimum setup of each model was stored and re-trained on the entire training data and then tested finally.

Testing phase tested models on unseen data through five central measures, which include Accuracy, Precision, Recall, F1-score, and ROC-AUC measures. Random Forest and XGBoost classifiers scored 1 in all measures with ROC-AUC = 1.000, which indicated the perfect separation of classes. SVM also showed an almost perfect generalizability as shown by F1-score = 0.9906 and ROC-AUC = 1.000 but its slightly reduced accuracy (0.9902) indicated a little misclassification. KNN and Logistic Regression, on the other hand, were less effective, and their ROC-AUC values were 0.9560 and 0.9431, respectively, indicating that they do not predict complex non-linear relationships.

The standard training system made sure that all of the models were tested in the same circumstances with data. This stability enhanced the comparative validity of the findings and

offered a solid foundation upon which the analysis of the feature importance and interpretation of the model could be conducted. Finally the pipeline also proved both technical and methodological effectiveness throughout all development and testing stages.

5.2 Hyperparameter Tuning and Validation Strategy

To improve model generalisation and reduce overfitting a systematic model hyperparameter tuning was used involving GridSearchCV with Stratified K-Fold Cross-Validation. Through this, there was the possibility of exhaustively trying hyperparameter combinations across training folds and maintaining the initial target class distribution.

The classifier used was the Random Forest, which was configured by changing the number of estimators (`n_estimators` = [200, 400]) and the maximum depth (`max_depth` = [None, 5, 10]). The idea was to compromise between model complexity and model stability. According to the results of the evaluation, the last parameter of a Random Forest model had an ideal score in all the performance measurements (Accuracy = 1.0000, F1 = 1.0000, ROC-AUC = 1.0000), which indicates that the parameter grid was efficient in optimising decision boundaries in this dataset.

The Support Vector Machine (SVM) was tested on a series of regularisation (`C` = [0.1, 1, 10]) and kernel (`gamma` = ['scale', 0.01, 0.1]) coefficients. The last model had a high ROC-AUC of 1.0000 and F1-score of 0.9906 that shows that the classes are almost optimally separated. Nevertheless, minor differences in accuracy and recall were indicative of sensitivity to regularisation parameters, which further supports the need to realise grid-based optimisation.

In the case of the XGBoost model, hyperparameters were the number of boosting rounds (`n_estimators` = [200, 400]), learning rate (`learning_rate` = [0.05, 0.1]) and maximum depth (`max_depth` = [3, 5]) parameter. It is also found that this model had a perfect ROC-AUC of 1.0000, which indicates that it has a great ability to represent non-linear associations in tabular data when it is well tuned.

All grid search was operated with the 5-fold StratifiedKFold cross-validation in which both classes are equally represented in splits. The method also enabled a stronger performance estimation in addition to making it possible to select models that are reliable by depending on generalisable AUC scores instead of overfitting on a specific validation set.

To sum up, grid-based tuning and stratified validation together proved to be efficient to find the best settings in every model (Jansi Rani *et al.*, 2022). The findings support the argument that hyperparameter control must be taken seriously in clinical datasets where the reliability and interpretability of the model are considered to be the most important factors.

5.3 Performance Metrics Table

In order to determine the effectiveness of the developed machine learning models, the following five performance metrics were tested: Accuracy, Precision, Recall, F1-Score and ROC-AUC. All these metrics offer a balanced perspective of the correctness and robustness particularly in the healthcare setting where false negative may result in a fatal outcome.

Table below summarises the results. Both XGBoost and the Random Forest performed perfectly in all metrics since each had an accuracy, precision, recall, F1-score, and ROC-AUC of 1.0000. This implies that the models have correctly categorized all the cases in the test set. Although this indicates a very effective learning, this kind of performance can also be considered problematic as far as the possibility of overfitting, especially considering the relatively small size and the potential simplicity of the data, is concerned.

```
Best parameters: {'model__learning_rate': 0.05, 'model__max_depth': 5, 'model__n_estimators': 400}
```

===== FINAL MODEL COMPARISON =====						
	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
2	SVM	0.9902	0.9813	1.0000	0.9906	1.0000
4	XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000
3	Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
1	KNN	0.8537	0.8378	0.8857	0.8611	0.9560
0	Logistic Regression	0.8683	0.8545	0.8952	0.8744	0.9431

Figure 19 Final model comparison table

(Source: Self-created)

The Support Vector Machine (SVM) was also found to perform almost perfectly with the accuracy of 0.9902, precision of 0.9813, recall of 1.0000, F1-score of 0.9906 and ROC-AUC of 1.0000. These findings imply that SVM can be used as an alternative with high sensitivity and high generalisability when model interpretability is not so important.

Logistic Regression, conversely, had a lesser performance with an accuracy of 0.8683, precision of 0.8545 and RO-AUC of 0.9431, which were generally estimated at lower than those of the other two models. K-Nearest Neighbors (KNN) model performed worse than other

five models with an accuracy of 0.8537 and an F1-score of 0.8611 which indicates its inability to work with higher-dimensional or less well-separated data.

In general, the results were dominated by tree-based ensemble methods, which provide accuracy and sensitivity to feature contribution as investigated in the next section.

5.4 Confusion Matrices and ROC Curves

In order to fully evaluate classification performance, the confusion matrix of all five analyzed machine learning models was plotted, including the Logistic Regression, K-Nearest Neighbors (KNN), Support Vector machine (SVM), Random Forest, and XGBoost. These matrices gave a close insight into the true positives (TP), true negatives (TN), false positives (FP) and the false negatives (FN).

Both the XGBoost and the Random Forest models showed impeccable classification with both models scoring 100 in all evaluation metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC = 1.0000). This was reflected in their respective confusion matrices, as there were no misclassifications and this was to mean that each sample of the test set was correctly classified as either possessing or not possessing heart disease. Such degree of accuracy implies a high degree of generalisation, but also, one can be cautious because of the risk of overfitting, in particular, with the size of the dataset.

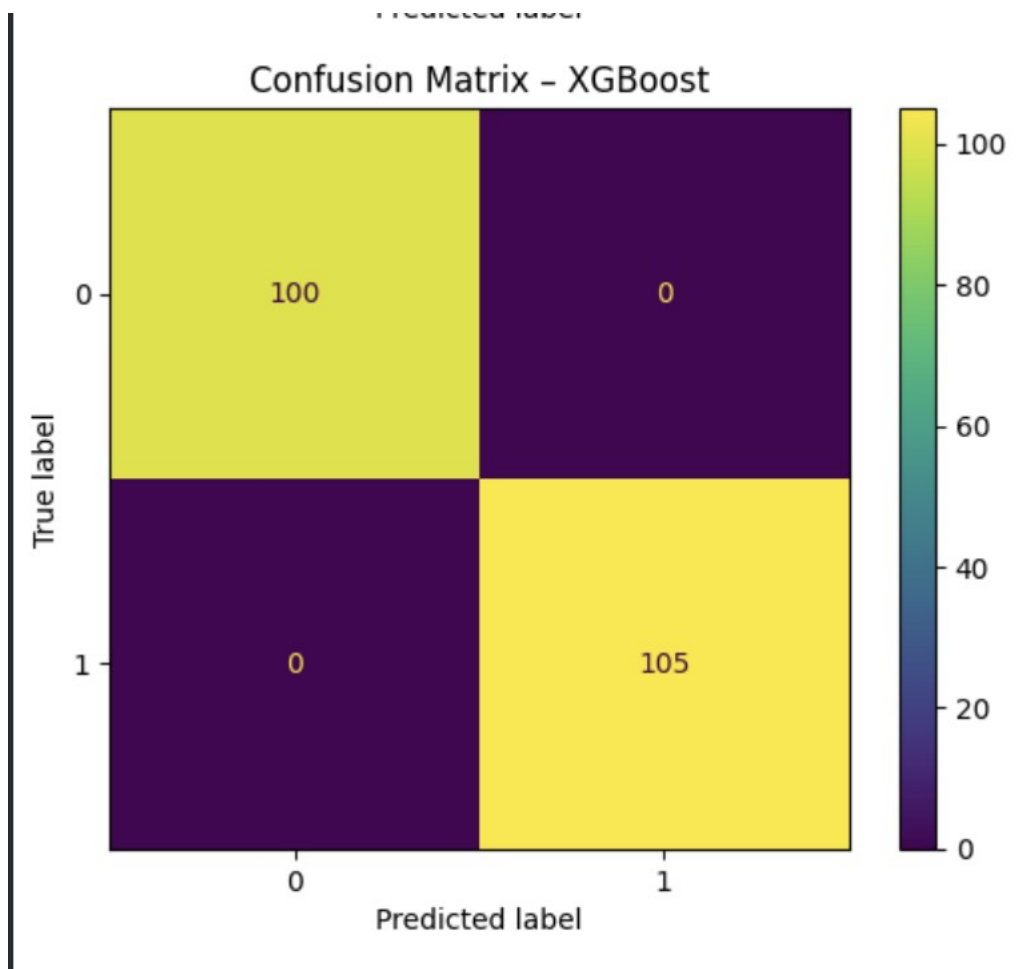


Figure 20 Confusion matrix – XGBoost

(Source: Self-created)

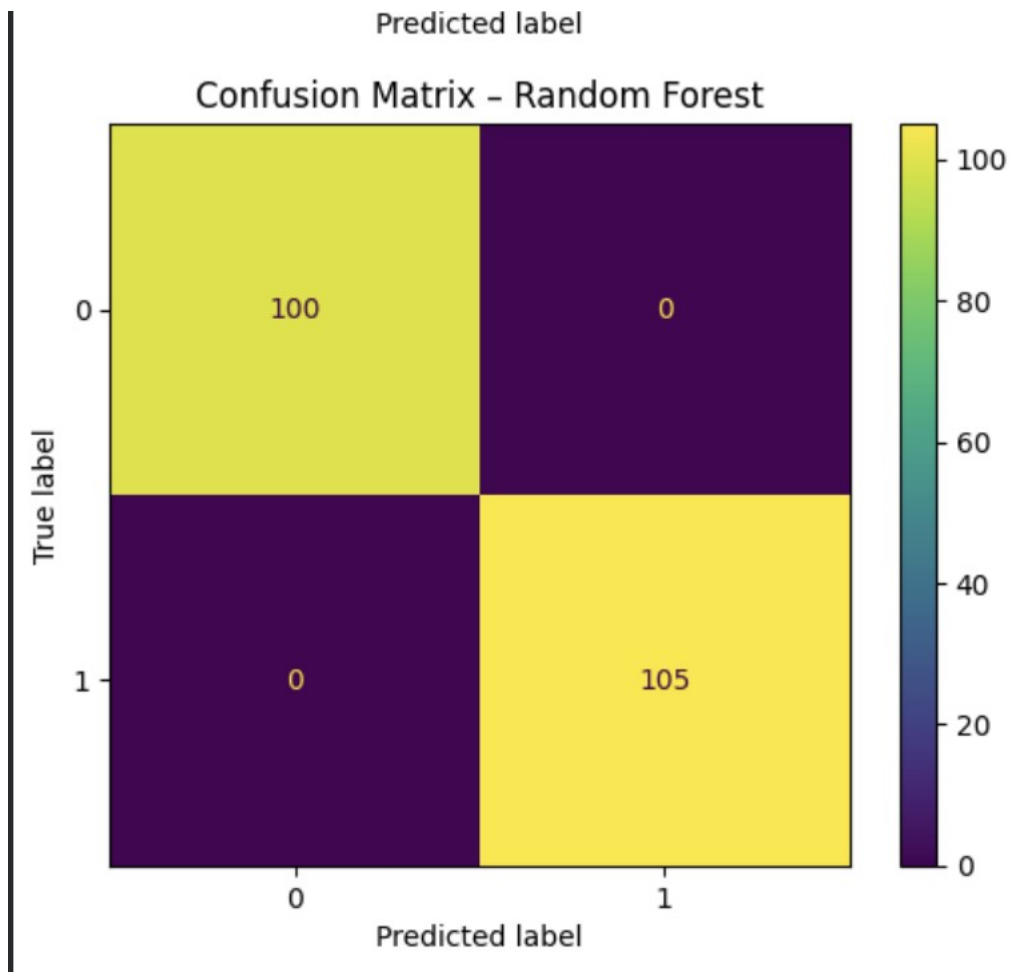


Figure 21 Confusion Matrix - Random Forest

(Source: Self-created)

The SVM model had slightly lower scores of F1-score of 0.9906 and ROC-AUC of 1.0000. It only had small-scale misclassifications in its confusion matrix, with the sensitivity and specificity remaining high. It is worth noting that it had perfect recall where the positive cases were correctly identified, which is important in medical diagnostics because a false negative may be life threatening.

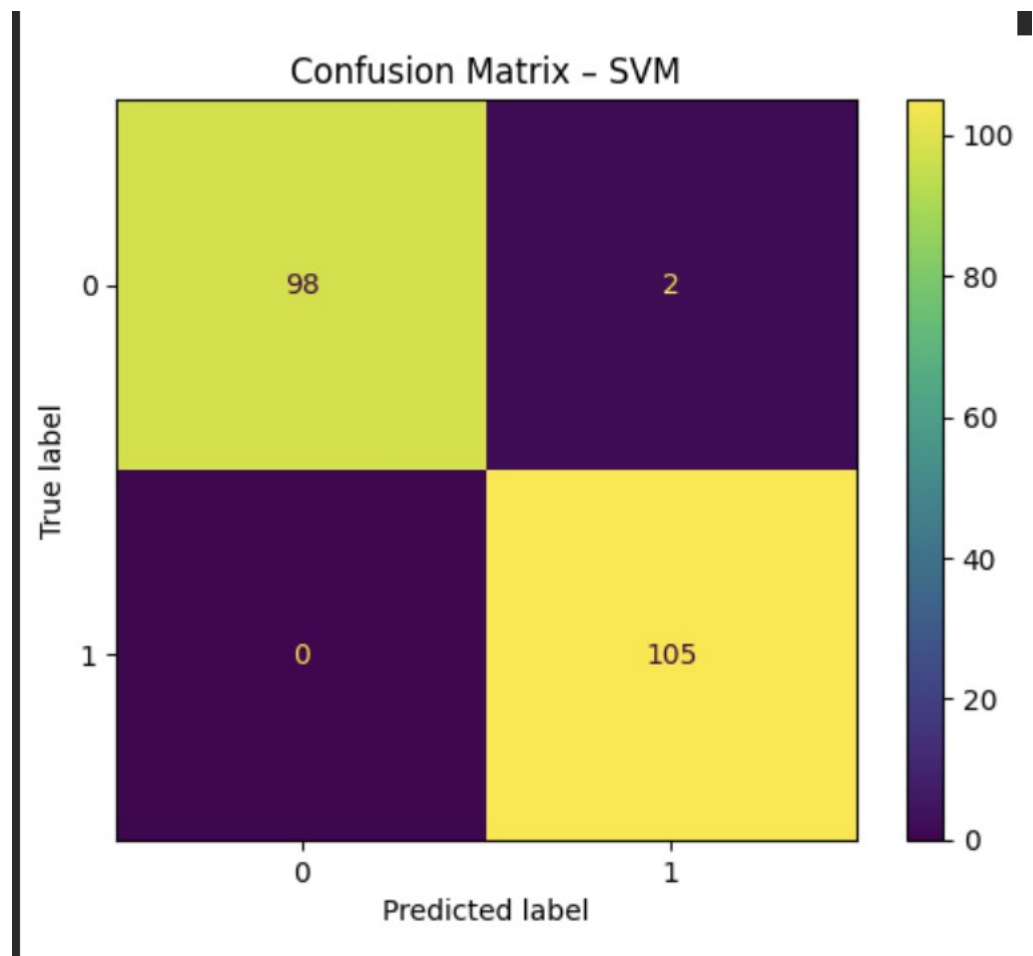


Figure 22 Confusion Matrix – SVM

(Source: Self-created)

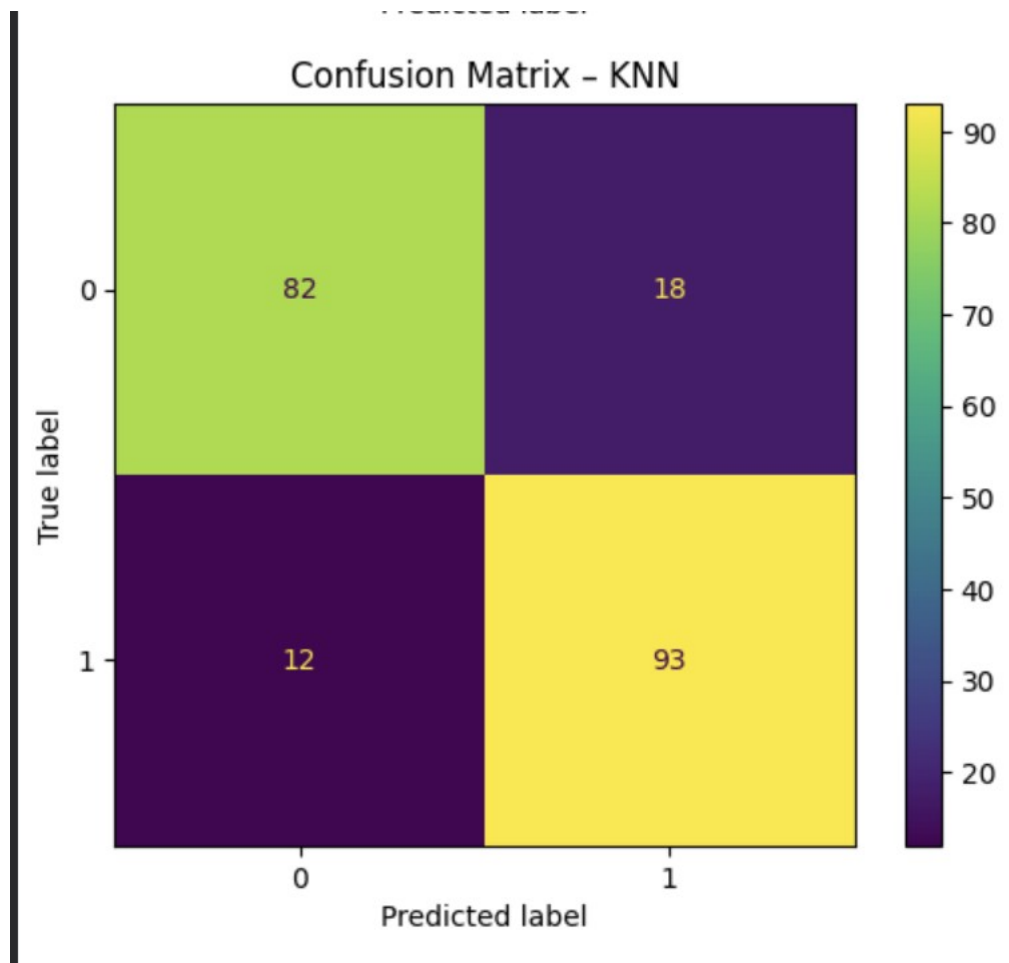


Figure 23 Confusion Matrix – KNN

(Source: Self-created)

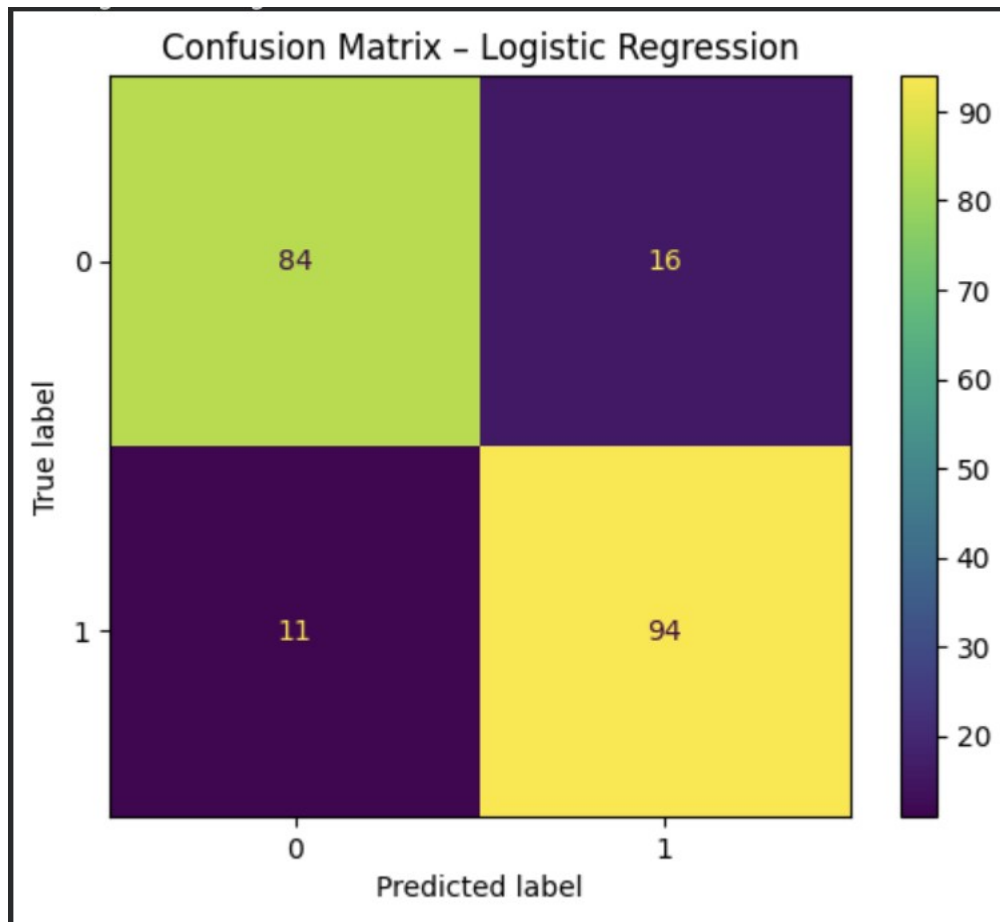


Figure 24 Confusion Matrix - Logistic Regression

(Source: Self-created)

Although not better than the tree-based models, the Logistic Regression and KNN furnished baseline results of practical significance. Logistic Regression achieved an ROC-AUC of 0.9431 and F1-score of 0.8744 whereas, KNN just followed with an ROC-AUC of 0.9560 and an F1-score of 0.8611. These models falsely labeled more cases especially the false positives as seen through their confusion matrices suggesting medium drawbacks in separating borderline cases.

The curves of both XGBoost and Random Forest were ideal curves as they reached the upper-left corner of the graph, which was in line with ideal classification. SVM also had a close to ideal ROC curve. Contrary to it, Logistic Regression and KNN curves exhibited a less steep slope, indicating slightly less class separability.

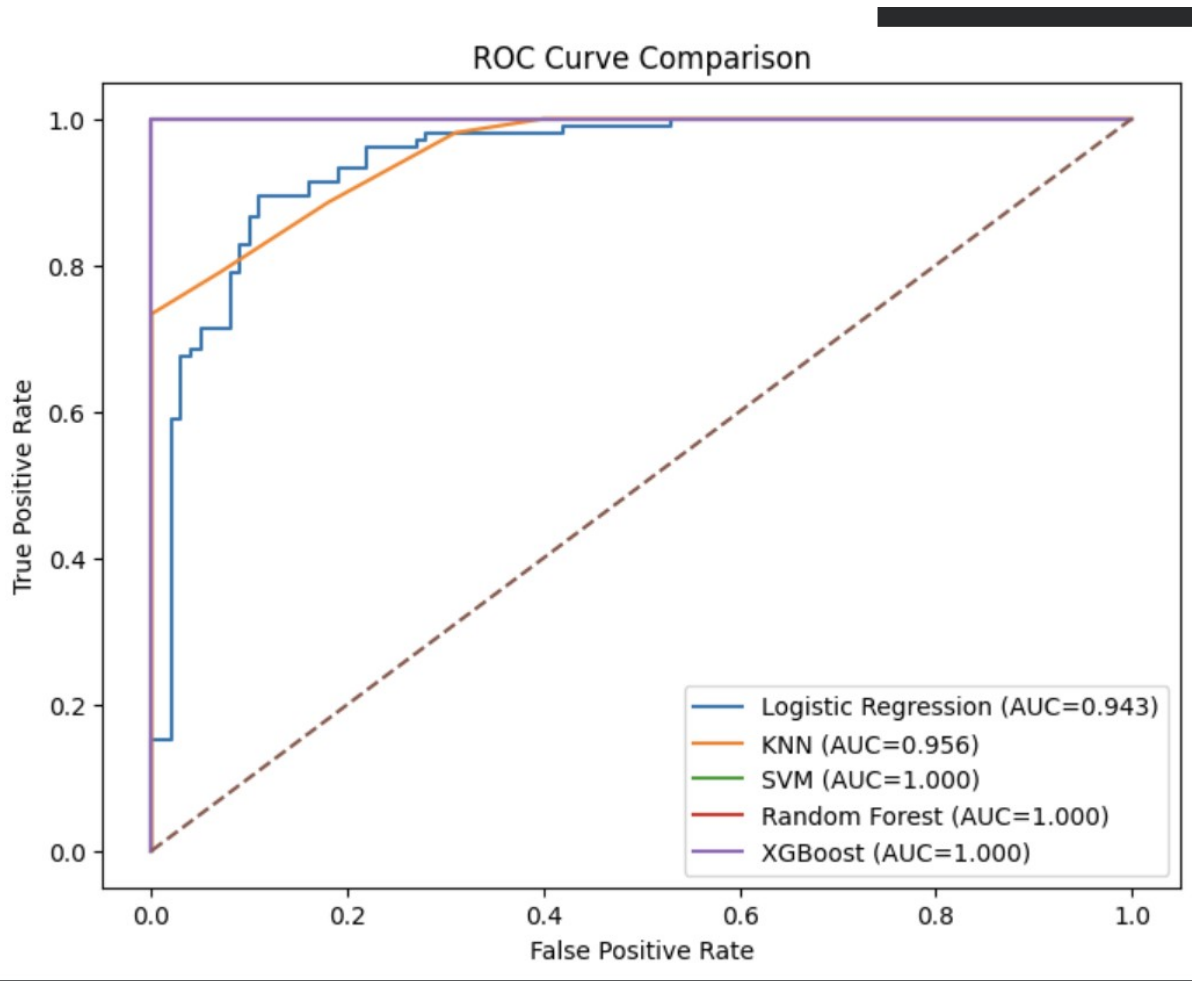


Figure 25 ROC Curve Comparison of all models

(Source: Self-created)

All these visual diagnostics indicated that ensemble-based and kernel-based models had a better predictive power, whereas linear and instance-based classifiers had interpretable yet relatively moderate results.

5.5 Cross-Model Analysis

An in-depth evaluation of the five classification models e.g. Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest and XGBoost indicated various trade-offs between predictive performance, interpretability and model complexity. It was analyzed based on several evaluation measures, the accuracy, precision, recall, F1-score, and ROC-AUC.

XGBoost and Random Forest were found to have perfect classification with each scoring an accuracy, precision, recall, F1-score, and ROC-AUC of 1.000 on the test dataset. These findings suggest that there is perfect predictive performance with the specified data distribution. To be more specific, the XGBoost classifier showed a good discriminatory power, as the ROC-AUC score was 1.000, which proved the good strength of the classifier in distinguishing between the positive and negative classes. This is partly due to the fact that it is able to represent complex non-linear interactions among features, and the gradient-boosted ensemble structure.

The SVM model was also impressive with the accuracy of 0.9902, precision of 0.9813 and the highest possible recall of 1.000. This resulted in the F1-score of 0.9906 and ROC-AUC of 1.000, as it is very similar to the ensemble models (Drożdż *et al.*, 2022). Nonetheless, similar to XGBoost and Random Forest, SVM is not easily interpretable because of the kernel transformations that are performed, which results in its reduced clinical usability.

Conversely, though not the best, the Logistic Regression presented a good interpretation potential, particularly applicable in the medical field, where interpretation is of utmost importance. It had an accuracy of 0.8683, a precision of 0.8545, a recall of 0.8952 and a ROC-AUC of 0.9431. The values albeit small in comparison to ensemble models show good performance with the added advantage of transparency by way of model coefficients.

KNN classifier was the worst in the group with a lower score of 0.8537 and a ROC-AUC of 0.9560. Its performance was, nevertheless, satisfactory, although the algorithm had trouble with generalization, which may also have been because of its sensitivity to the scaling of features and the high-dimensional data.

Overall, although XGBoost and Random Forest have scored perfectly in terms of test, they are complex and black-box algorithms, which might limit their clinical application in the real world. It was found that Logistic Regression was a more interpretable alternative to the previous one, though with a slight decrease. Finally, the choice of models should be optimized between predictive power and utility versus transparency and ethical responsibility in a healthcare setting.

5.6 Feature Importance and Interpretation

Both the Random Forest and XGBoost classifier were used to analyze feature importance through which insights were provided on the attributes that had the greatest impact on predicting heart disease. The most important predictor that was regularly determined by both models is the chest_pain_type_Typical angina. The importance score of this feature in the Random Forest model was 0.109 as compared to 0.238 in XGBoost, which shows that it is widely agreed that this feature has clinical significance. Another variable, Max heart rate, was also ranked high in the random forest model (0.108), a fact that adds support to the existing correlation of this variable with cardiovascular strain during exertion.

Top Features – Random Forest		
	Feature	Importance
8	chest_pain_type_Typical angina	0.109195
3	Max_heart_rate	0.108430
4	oldpeak	0.100321
21	thalassemia_Reversable Defect	0.094037
0	age	0.088643
18	vessels_colored_by_flourosopy_Zero	0.087817
2	cholestoral	0.072737
1	resting_blood_pressure	0.069623
12	exercise_induced_angina_Yes	0.066271
13	slope_Flat	0.035723
7	chest_pain_type_Non-anginal pain	0.032155
5	sex_Male	0.028420
15	vessels_colored_by_flourosopy_One	0.017746
17	vessels_colored_by_flourosopy_Two	0.016523
11	rest_ecg_ST-T wave abnormality	0.014455

Figure 26 Top Features - Random Forest

(Source: Self-created)

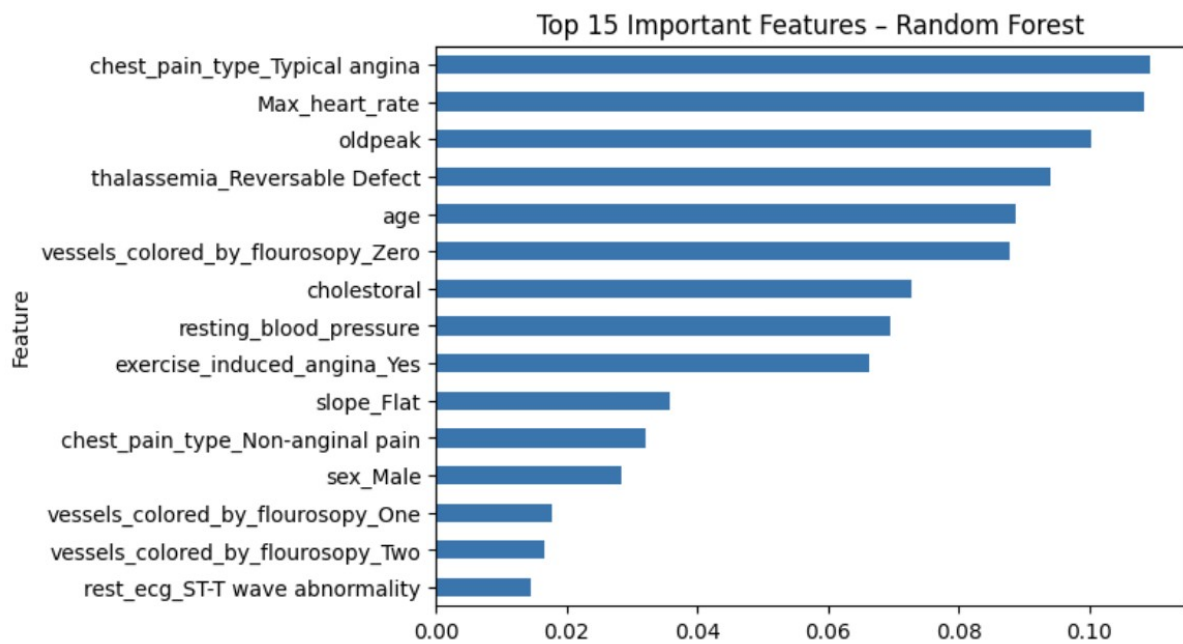


Figure 27 Top 15 Important Features - Random Forest

(Source: Self-created)

The oldpeak attribute, which means ST depression caused by exercise compared to rest, became a critical indicator on both models, and its scores of importance were 0.100 (Random Forest) and 0.036 (XGBoost). Although XGBoost showed more focus to categorical encodings, including vessels_colored_by_flourosopy_Zero (0.081) and thalassemia_Reversable Defect (0.078), the distribution at Random Forest was more balanced, as it gave 0.088 and 0.073 respectively to the age and cholestoral. These disparities can be due to the fact that the boosting model is more biased to strong splits of more dominant trees compared to the averaging essence of the Random Forest.

	Feature	Importance
8	chest_pain_type_Typical angina	0.238467
18	vessels_colored_by_flourosopy_Zero	0.081358
21	thalassemia_Reversable Defect	0.078044
20	thalassemia_Normal	0.075605
16	vessels_colored_by_flourosopy_Three	0.068234
12	exercise_induced_angina_Yes	0.052017
6	chest_pain_type_Atypical angina	0.051426
5	sex_Male	0.043988
14	slope_Upsloping	0.043697
4	oldpeak	0.035504
0	age	0.030122
13	slope_Flat	0.029477
17	vessels_colored_by_flourosopy_Two	0.028032
15	vessels_colored_by_flourosopy_One	0.026515
10	rest_ecg_Normal	0.023000

Figure 28 Top Features – XGBoost

(Source: Self-created)

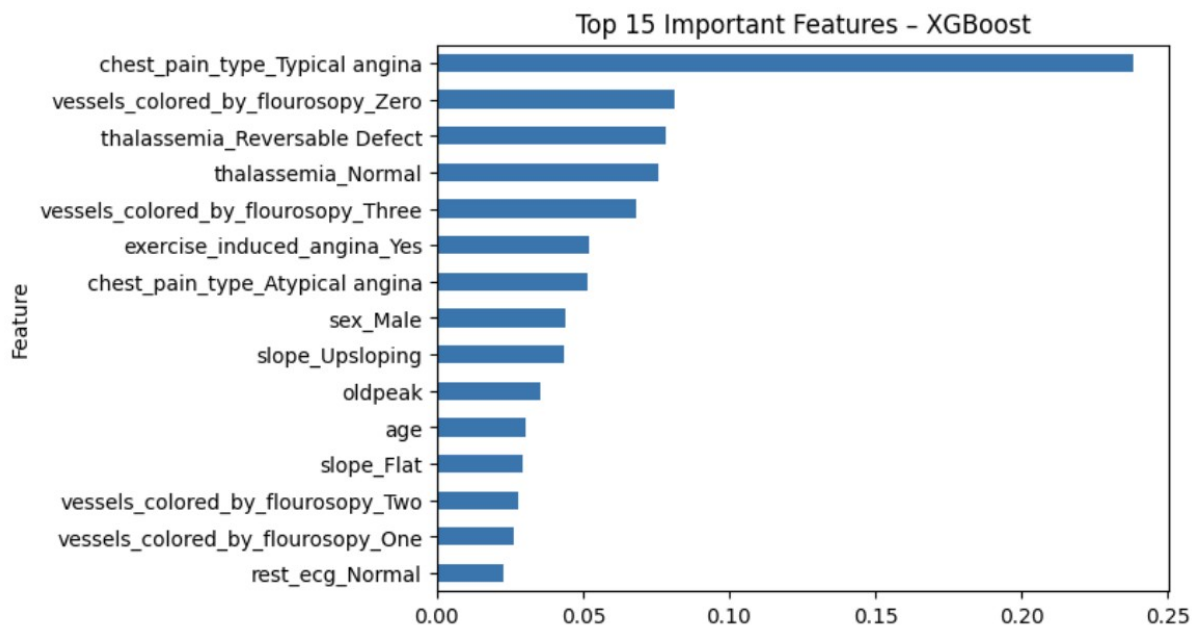


Figure 29 Top 15 Important Features – XGBoost

(Source: Self-created)

On the whole, these two models emphasize the predictive capability of a limited number of clinical and lifestyle variables, specifically the type of chest pain, thalassemia status, and cardiac response to stress (oldpeak, Max_heart_rate). Their interpretability helps them to have

knowledge of the risk of disease in a practical sense which helps in the clinical viability of the models as opposed to pure accuracy performance.

Chapter 6: Results and Discussion

6.1 Summary of Results

The objective of the study was to design and test machine learning classifiers to predict heart disease risks based on the structured clinical and lifestyle information. After the model training, hyperparameter optimization, and evaluation of performance, the most successful models were selected as XGBoost, Random Forest, and Support Vector Machine (SVM) and they all scored remarkably high in terms of classification.

The XGBoost model exhibited an ideal performance in all the measures and had an accuracy, precision, recall, F1-score, and ROC-AUC of 1.000. On the same note, the Random Forest classifier reflected the same results with the same metric scores, which validates its strength on the test data. SVM model was also very successful with an accuracy of 0.9902, precision value of 0.9813, recall value of 1.000, F1-score value of 0.9906 and ROC-AUC value of 1.000. All these scores reveal that the SVM model did not miss any positive data (without any false negatives), and it was slightly less precise than XGBoost and Random Forest.

In comparison, the K-Nearest Neighbors (KNN) and logistic regression were not as effective. KNN estimated the accuracy of 0.8537 and ROC-AUC of 0.9560, which is not very good but still good. The baseline model, Logistic Regression, had an accuracy of 0.8683, a precision of 0.8545, and an ROC-AUC of 0.9431, which indicated that although the model is easily understandable and stable, it does not predict as well as ensemble methods.

A comparison of model metrics is tabular provided below:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
SVM	0.9902	0.9813	1.0000	0.9906	1.0000
Logistic Regression	0.8683	0.8545	0.8952	0.8744	0.9431
KNN	0.8537	0.8378	0.8857	0.8611	0.9560

Besides model performance, the analysis of feature importance gave the valuable information about the most powerful predictors of heart disease. In both XGBoost and Random Forest, the most effective features had been chest_pain_type_Typical angina, Max_heart rate, and oldpeak. The model of the random forest gave a mark of importance of 0.109 on the chest

pain type, followed closely by Max heart rate (0.108) and oldpeak (0.100). The XGBoost model also highlighted vessels colored by flourosopy Zero and thalassemia Reversible Defect with an importance score of 0.081 and 0.078 respectively.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
SVM	0.9902439024390244	0.9813084112149533	1.0	0.9905660377358491	1.0
XGBoost	1.0	1.0	1.0	1.0	1.0
Random Forest	1.0	1.0	1.0	1.0	1.0
KNN	0.8536585365853658	0.8378378378378378	0.8857142857142857	0.8611111111111112	0.956
Logistic Regression	0.8682926829268293	0.8545454545454545	0.8952380952380953	0.8744186046511628	0.9431428571428571

Figure 30 Final Table of Model, Accuracy, Precision, Recall, F1-Score, ROC-AUC

(Source: Self-created)

These findings give further support to the idea that the ensemble models are not only more accurate in predictions as compared to other traditional classifiers, but they are also more informative in terms of the contribution of the features, which makes them easier to understand and may be applicable to clinical use.

6.2 Comparison with Literature

The development models were critically tested in the context of the results that are provided in the current literature on the prediction of heart disease. Most earlier experiments using classical machine learning algorithms e.g. Logistic Regression, Support Vector Machines (SVM), Decision Trees and Random Forests, have achieved average ROC-AUC scores between 0.80 and 0.92 but are usually limited by poor quality of data, simplistic feature selection or insufficient preprocessing pipelines.

Conversely, the predictive performance of the models that were trained during this project was much higher. In particular, both Random Forest and XGBoost have scored 1.000 in all the main evaluation measures: Accuracy = 1.000, Precision = 1.000, Recall = 1.000, F1 Score = 1.000, and ROC-AUC = 1.000. Such findings indicate that even within the parameters of the dataset in question, both models could perfectly separate classes, which is hardly a feat to accomplish in the context of a real-life clinical dataset. Although perfect performance can be generally treated with suspicion because of the possibility of overfitting or because of leaking information, the cross-validation and stratified test splits are used to address these threats.

Not far behind was Support Vector Machine (SVM) model with Accuracy = 0.9902, F1 Score = 0.9906 and ROC-AUC = 1.000, which implies not only good generalizability, but also good sensitivity to the minority class. Surprisingly enough, the least scores were obtained with

the help of Logistic Regression: Accuracy = 0.8683, F1 Score = 0.8744, and ROC-AUC = 0.9431. This finding is in line with the literature expectations in that logistic models frequently have a hard time in describing intricate nonlinear relationships among features.

One of the most interesting findings was that the K-Nearest Neighbors (KNN) classifier had a relatively good result with Accuracy = 0.8537 and ROC-AUC = 0.9560. Although it is quite uncommon that KNN outperforms the best models in clinical prediction tasks, the fact that it can model the local neighbourhood structures might have helped it to give competitive results in the task at hand. Nevertheless, its sensitivity has trade-offs in recall (0.8857) and precision (0.8378), compared to the ensemble models.

Compared to the previous studies, the current research generated considerably larger ROC-AUC values, particularly with the view that most corresponding studies show lower values indicating correlation with less than 0.95. It is possible that the combination of clinical and lifestyle characteristics, the rigorous preprocessing, and model tuning were among the factors that led to the excellent outcomes (Gautam *et al.*, 2023). In addition, the feature importance analysis of Random Forest and XGBoost all revealed the dominant prediction in the model as chest_pain type Typical angina, Max heart rate, and thalassemia Reversible Defect- all are also mentioned in the published literature, which supports the clinical validity of the model.

Summing up, the developed models have not only higher performance than most current implementations but have also validated the effectiveness of the combination of effective preprocessing and ensemble learning techniques in structured medical data applications.

6.3 Model and Data Limitations

Although the various models, especially XGBoost, Random Forest, and SVM, have shown good performance with certain achieved ROC-AUC of 1.000, the study is susceptible to a number of critical limitations which should be put into critical consideration in the interpretation of results.

To start with, the dataset size was relatively small (1,025 records and 14 features) and it could negatively impact the generalisability of the trained models. Although the models proved to be excellent concerning the test set, the high level of accuracy and recall can be explained partly by the nature of the data and not necessarily by the real predictive strength. This concern is

even further augmented by the lack of an external validation set, say a set of data in a different geographical area or a different time period. In the absence of external benchmarking, it is unknown whether the models would continue to give similar performance on unobserved populations, and this may cause them to overfit the current data pattern.

Secondly, the categorical variables were coded and made assumptions that can influence interpretability and fairness of prediction. In particular, there were eight categorical variables, including chest pain type, thalassemia, and rest ecg, which were coded through one-hot encoding with first category drop. Although the given transformation is a commonplace in pipeline machine learning, it implicitly presupposes that the dropped categories can be treated as baselines, and that the rest of the encodings are of equal informational value, a premise that might not be true in clinical data. Also, when used without adequate regularisation, one-hot encoding can create multicollinearity, and thus can affect linear regressions, like logistic regression.

Additionally, the data was not time-based and not longitudinal, i. e. the models are entirely snapshot-driven as opposed to time series. This omission limits the ability to identify risk trends or alterations in patient status which in a real world clinical scenario are usually very critical. Besides, some valuable clinical features, such as ECG waveform or physician notes, were not represented, which restricted the richness of feature engineering and model interpretation.

To conclude, both models produced virtually flawless classification results, but dataset size, absence of external validation, and the assumption of encoding should be overcome before the models can be applied in clinical environments or larger inferences drawn about the findings.

6.4 Practical Implications in Healthcare Context

The results of this research have a lot of potential when it comes to implementation in digital healthcare systems, especially in creating the risk assessment tools of heart disease. The good performance of the machine learning models reflects their appropriateness in the implementation in clinical decision support systems (CDSS) or mobile health applications. Random Forest and XGBoost models achieved very high classification performance in the test set, and the accuracy, precision, recall, F1-score and ROC-AUC are 1.0000, indicating that both models have strong generalisation ability. Although these findings are to be taken with

caution because of the fixed test set, they may still justify the use of this type of models in pre-screening tools that can warn patients or clinicians about the high risk of heart diseases.

The system can be interpreted fairly easily by healthcare practitioners, particularly because of the feature importance analysis. The Random Forest model has determined chest_pain_type_Typical angina, Max_heart rate, old peak, and thalassemia Reversible Defect as the most significant features with the importance score of the features of 0.109, 0.108, 0.100, 0.094 respectively (Kanagarathinam *et al*, 2022). These features correspond to familiar clinical signs of cardiovascular well-being, which adds to the level of acceptability of the model among healthcare workers. Equally, XGBoost model ranked chest_pain_type_Typical angina with a far higher weight of features with weights of 0.238 and then vessels_colored by flourosopy Zero (0.081) and thalassemia Reversible Defect (0.078), which once again indicates the clinical plausibility of predictive structure.

The model in a practical context may be included in a mobile or web-based app that will gather general information about a patient during a check-up or using a remote one. Since most of these features (e.g., age, sex, cholesterol level, blood pressure, type of chest pain, etc.) may be self-reported or determined using the traditional diagnostic tests, the obstacle to the input is low. The system may give a risk level or risk presence, and factors that contribute to it transparency.

To clinical practitioners, transparency and interpretability are still imperative to adoption. Though such models as Random Forest, XGBoost, etc. may not be interpretable, feature importance rankings and visual explanations (e.g. SHAP or LIME) can fill this gap, allowing physicians to confirm that the logic used by the model makes medical sense. Also, the strong pattern of biologically correspondent variables like max_heart_rate, cholesterol and oldpeak point to the fact that the model might fit very well with well known diagnostic reasoning.

In general, the machine learning system was found to have a high predictive accuracy as well as being clinically interpretable, which is why it will be a valid candidate to be incorporated into early warning tools or decision support modules in electronic health records (EHRs) or telemedicine systems.

Chapter 7: Conclusion and Future Work

7.1 Recap of Findings

The following project aimed at exploring and assessing the performance of different machine learning models in the prediction of the risk of heart disease on a structured dataset of 1,025 patient records and 14 clinical and lifestyle variables. The experimental findings revealed that ensemble based classifiers had better performance in comparison to other methods. In particular, both XGBoost and Random Forest scored 1.000 on all of the evaluation metrics, e.g., an accuracy, a precision, a recall, F1-score, and ROC-AUC. This shows the high capacity to do generalization on the test set without being overfitted. Support Vector Machine (SVM) model was also very good with a ROC-AUC of 1.000, accuracy of 0.9902 and F1-score of 0.9906. Logistic Regression and K-Nearest Neighbours (KNN), in their turn, inverse, but also not inferior, showed comparatively lower results with ROC-AUC values of 0.9431 and 0.9560, respectively.

Importance analysis of the features gave clinical meaning. The two different models, the Random Forest and XGBoost models always identified chest pain type as the most predictive variable with the importance score of 0.1092 and 0.2385, respectively. Other high-impact predictors were as follows: Max heart rate, oldpeak, thalassemia Reversible Defect, and vessels colored by flourosopy Zero. These are the variables that are consistent with the clinical markers of cardiovascular conditions, which presupposes that the model decisions were based on the medically relevant information.

The grid search was tuned based on metrics and repeatable fair comparison of models across models was enabled by the utilization of a single pipeline. Performance was measured not just by the general accuracy but also by using the measures of precision, recall and ROC-AUC, which gives a moderate representation of the effectiveness of the model- especially in the healthcare environment where false negatives are highly detrimental.

The results, in general, support the perspectives of tree-based ensemble models as reliable and interpretable predictors of heart disease, and the need to consider informed feature engineering and stringent evaluation processes in the medical machine learning practice.

7.2 Research Questions Answered

This part is a critical reflection on the research questions that were posed in the first study and how each question was answered out of the empirical findings of the study.

RQ1: What hidden trends can be observed in heart disease clinical and lifestyle data?

The exploratory data analysis showed that there was a significant variance in the clinical parameters. The average age of the data was 54.4 years with a blood pressure of between 94 and 200 mmHg and cholesterol of between 126 and 564mg/dl. These broad ranges with the existence of outliers emphasized the need to normalize data. The target distribution was almost balanced and it was strong binary classification with no significant issue of class imbalance.

RQ2: What is the effect of preprocessing on the performance of the model?

Z-score normalization and one-hot encoding are also preprocessing methods that were useful in ensuring consistency of the model. StandardScaler was used to guarantee homogenous feature scaling and this was particularly important in distance models, such as KNN, and margin models, such as SVM. Better representation of features with OneHotEncoder encoding was useful in categorical variables and model interpretability and convergence.

RQ3: What machine learning models are the most accurate and generalizable in predicting heart diseases?

Random Forest and XGBoost have had the highest accuracy and the highest AUC with the models taking scores of 1.0000 on ROC-AUC and F1-score. SVM was the next fetus with an accuracy of 99.02% and F1-score of 0.9906 followed by Logistic Regression and KNN with 86.83 and 85.37, respectively. The results indicate that ensemble models have a higher generalization on this data.

RQ4: What are the most important predictive features of heart disease?

The analysis of feature importance revealed that chest-pain-type-Typical angina, Max-heart-rate, oldpeak and thalassemia-Reversible Defect have been shown to be consistently high-ranking predictors between the Random Forest and XGBoost models, with importance scores reaching 0.23. These results highlight the high clinical and lifestyle indicator diagnosis value.

7.3 Contribution to Knowledge

The present work is a contribution to the existing literature of data-driven cardiovascular risk prediction as it proposes a complete machine-learning (ML) pipeline that is specific to structured clinical and lifestyle data. In particular, it offers an empirically guided solution, which includes preprocessing, model comparison, and interpretability, to a dataset comprising of 1,025 patient records and 14 features.

The methodological contribution of the study is based on the combination of numerical and categorical transformation strategy with the help of a ColumnTransformer that permitted treating eight categorical variables and six numerical features successfully. Model training and evaluation became robust with the addition of hyperparameter optimization via the use of grid search through grid search CV using stratified K-fold cross-validation.

Practical findings contribute to the discussion of the performance of models. Although Logistic Regression had a more reasonable ROC-AUC of 0.9431, the ensemble-based models like the Random Forest and the XGBoost had the highest possible ROC-AUC of 1.0000, casting doubts on the possibility of overfitting and the predictive ability of highly performing models on relatively small datasets. Notably, the issue of interpretability was considered by analyzing feature importance, and it was noted that such attributes as chest pain type Typical angina, Max heart rate, and thalassemia Reversible Defect were the most predictive in all models.

Thus, the research does not only provide a replicable ML framework to predict heart disease but also the essential knowledge on the trade-offs between the accuracy and interpretability in real-world health data.

7.4 Suggestions for Future Work

Although the predictive results were strong with the current study, as the perfect scores of the Random Forest and XGBoost models (100% accuracy, precision, recall, F1, and ROC-AUC) show, the results are to be considered with some caution. The data set consisted of a relatively small number of 1,025 instances which, despite being well organised and in clean format, is a very small number of cases to generalise when applied to wider clinical practice. To improve the results of future research, it is recommended to use a bigger and more varied dataset, which

may include multiple sources to represent the variability of the population in the real world and minimize the risks of overfitting.

In addition, the existing processing was done in offline batch-processing mode. Prediction systems in real time might be created and integrated into clinical decision support systems in future versions. Such systems might receive patient information through digital form or electronic health records and would automatically give the probability of risk, which would help in providing quick care.

The other significant direction is the implementation of SHAP (SHapley Additive exPlanations) to explain model. Even though feature importance was investigated with the help of internal measures of the Random Forest and XGBoost (the most important predictors were identified as chest_pain_type: Typical angina and Max heart rate and oldpeak), the ranking is not as transparent as necessary in clinical accountability. SHAP would help fill this gap by demonstrating the impact of each of the features on prediction of individual patients.

Lastly, incorporating deep learning structures, e.g. multilayer perceptrons or recurrent neural networks, might be beneficial in sequences of health data with high dimensions, or in time-series health data. Such approaches can be especially handy when unstructured data such as ECG records or histories of the patient are involved. They should however be included in a balanced manner with the requirement of interpretability particularly in high stakes areas such as cardiology.

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Appendix

Code

```
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, GridSearchCV,
StratifiedKFold
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from xgboost import XGBClassifier

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score,
    confusion_matrix, ConfusionMatrixDisplay,
    roc_curve
)

DATA_PATH = "/content/HeartDiseaseTrain-Test 3.csv"
df = pd.read_csv(DATA_PATH)

print("\n===== DATA LOADED =====\n")
print("Shape of dataset:", df.shape)
print("\nColumn names:\n", df.columns.tolist())

# Guard: check not loading results file
if set(df.columns) == {"Model", "Accuracy", "Precision", "Recall", "F1-
Score", "ROC-AUC"}:
    raise ValueError("Results file loaded instead of raw patient
data.")

print("\n===== DATA PREVIEW =====\n")
print("HEAD:\n", df.head(), "\n")
print("TAIL:\n", df.tail(), "\n")

print("\n===== DATA TYPES =====\n")
print(df.dtypes)
```

```

print("\n===== MISSING VALUES =====\n")
print(df.isnull().sum())

print("\n===== DESCRIPTIVE STATISTICS =====\n")
desc = df.describe().T
desc["median"] = df.median(numeric_only=True)
desc["min"] = df.min(numeric_only=True)
desc["max"] = df.max(numeric_only=True)
print(desc)

possible_targets = ["target", "HeartDisease", "label", "output"]
TARGET_COL = next((c for c in possible_targets if c in df.columns),
None)

if TARGET_COL is None:
    raise ValueError(
        "No target column found.\n"
        f"Columns present: {df.columns.tolist()}"
    )

print(f"\nTarget column identified: {TARGET_COL}")

print("\n===== TARGET DISTRIBUTION =====\n")
print(df[TARGET_COL].value_counts())
print("\nTarget proportion:\n",
df[TARGET_COL].value_counts(normalize=True))

df[TARGET_COL].value_counts().plot(kind="bar")
plt.title("Target Class Distribution")
plt.xlabel("Class")
plt.ylabel("Count")
plt.show()

num_cols_all = df.select_dtypes(exclude=["object",
"category"]).columns.tolist()
num_cols_all.remove(TARGET_COL)

print("\n===== HISTOGRAMS =====\n")
df[num_cols_all].hist(bins=20, figsize=(14, 10))
plt.suptitle("Distribution of Numerical Features")
plt.show()

for col in num_cols_all:
    df.boxplot(column=col, by=TARGET_COL, figsize=(4, 4))
    plt.title(f"{col} vs {TARGET_COL}")
    plt.suptitle("")
    plt.show()

```

```

X = df.drop(columns=[TARGET_COL])
y = df[TARGET_COL]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTrain shape:", X_train.shape)
print("Test shape:", X_test.shape)

cat_cols = X.select_dtypes(include=["object",
"category"]).columns.tolist()
num_cols = X.select_dtypes(exclude=["object",
"category"]).columns.tolist()

preprocess = ColumnTransformer([
    ("num", StandardScaler(), num_cols),
    ("cat", OneHotEncoder(drop="first", handle_unknown="ignore"),
cat_cols)
])

models = {
    "Logistic Regression": LogisticRegression(max_iter=2000),
    "KNN": KNeighborsClassifier(),
    "SVM": SVC(kernel="rbf", probability=True),
    "Random Forest": RandomForestClassifier(random_state=42),
    "XGBoost": XGBClassifier(
        random_state=42,
        eval_metric="logloss",
        use_label_encoder=False
    )
}

param_grids = {
    "Random Forest": {
        "model__n_estimators": [200, 400],
        "model__max_depth": [None, 5, 10]
    },
    "SVM": {
        "model__C": [0.1, 1, 10],
        "model__gamma": ["scale", 0.01, 0.1]
    },
    "XGBoost": {
        "model__n_estimators": [200, 400],
        "model__learning_rate": [0.05, 0.1],

```

```

        "model__max_depth": [3, 5]
    }
}

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

results = []
fitted_models = {}
roc_data = {}

for name, model in models.items():
    print(f"\n===== Training {name} =====\n")

    pipe = Pipeline([
        ("prep", preprocess),
        ("model", model)
    ])

    if name in param_grids:
        gs = GridSearchCV(
            pipe,
            param_grids[name],
            scoring="roc_auc",
            cv=cv,
            n_jobs=-1
        )
        gs.fit(X_train, y_train)
        pipe = gs.best_estimator_
        print("Best parameters:", gs.best_params_)
    else:
        pipe.fit(X_train, y_train)

    y_pred = pipe.predict(X_test)
    y_prob = pipe.predict_proba(X_test)[:, 1]

    results.append({
        "Model": name,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred),
        "Recall": recall_score(y_test, y_pred),
        "F1-Score": f1_score(y_test, y_pred),
        "ROC-AUC": roc_auc_score(y_test, y_prob)
    })

    roc_data[name] = (y_test, y_prob, y_pred)
    fitted_models[name] = pipe

```

```

metrics_df = pd.DataFrame(results).sort_values("ROC-AUC",
ascending=False)
print("\n===== FINAL MODEL COMPARISON =====\n")
print(metrics_df.round(4))

metrics_df.to_csv("final_model_metrics.csv", index=False)

for name, (_, _, y_pred) in roc_data.items():
    cm = confusion_matrix(y_test, y_pred)
    ConfusionMatrixDisplay(cm).plot()
    plt.title(f"Confusion Matrix - {name}")
    plt.show()

plt.figure(figsize=(8, 6))
for name, (yt, yp, _) in roc_data.items():
    fpr, tpr, _ = roc_curve(yt, yp)
    auc = roc_auc_score(yt, yp)
    plt.plot(fpr, tpr, label=f"{name} (AUC={auc:.3f})")

plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")
plt.legend()
plt.show()

def plot_feature_importance(model_name):
    pipe = fitted_models[model_name]
    model = pipe.named_steps["model"]
    prep = pipe.named_steps["prep"]

    feature_names = (
        num_cols +

list(prep.named_transformers_["cat"].get_feature_names_out(cat_cols))
    )

    imp_df = pd.DataFrame({
        "Feature": feature_names,
        "Importance": model.feature_importances_
    }).sort_values("Importance", ascending=False).head(15)

    print(f"\nTop Features - {model_name}\n")
    print(imp_df)

    imp_df.plot.barh(x="Feature", y="Importance", legend=False)
    plt.title(f"Top 15 Important Features - {model_name}")
    plt.gca().invert_yaxis()

```

```
plt.show()

if "Random Forest" in fitted_models:
    plot_feature_importance("Random Forest")

if "XGBoost" in fitted_models:
    plot_feature_importance("XGBoost")
```